This 2026 version is the same as originally published in 2020 except Table 8 Version 3 has superseded Table 8 Version 1 & 2.

SD-STD-01.01, Revision H
Originally published June 16, 2020
Version 2 released October 24, 2024
Version 3 released April 2026

(Supersedes all earlier revisions of SD-STD-01.01 & SD-STD-01.02)

CERTIFICATION STANDARD
FORCED ENTRY AND BALLISTIC RESISTANCE OF BUILDING SYSTEMS

PHYSICAL SECURITY DIVISION
OFFICE OF PHYSICAL SECURITY PROGRAMS
BUREAU OF DIPLOMATIC SECURITY
U.S. DEPARTMENT OF STATE
WASHINGTON, D.C. 20520

BUREAU OF DIPLOMATIC SECURITY
UNITED STATES DEPARTMENT OF
STATE WASHINGTON, D.C.

CERTIFICATION STANDARD

FORCED ENTRY AND BALLISTIC RESISTANCE
OF BUILDING SYSTEMS

October 2024

Approved:

Mr. David J. Sparrowgrove
Director, Physical Security Programs, DS/C/PSP

BYRON X. BACIK

Byron Bacik
Division Chief, Physical Security Division, DS/PSP/PSD

Alec Williamson

Alec Williamson (acting)
Section Chief, Physical Security Research and Development, DS/PSD/RD

BRIEF TABLE OF CONTENTS

COMPLETE TABLE OF CONTENTS

PREFACE

MAJOR REVISIONS TO REVISION H

There have been a number of changes incorporated into this revision of the U.S. Department of State forced entry and ballistic resistant testing specification. This section provides a general overview.

Administrative and Procedural

- The latest revision to the test specification codifies many of the administrative and procedural processes the Bureau of Diplomatic Security (DS) has been following in recent years.
- Complete and detailed drawings are required of systems considered for subsequent testing and certification.
- The requirement that DS-approved systems must also comply with all applicable requirements and specifications promulgated by the US Department of States, Overseas Buildings Operations (OBO) Bureau is reinforced.
- Prescriptive requirements for window and door size and mounting gap are removed.
- The allowable size increase of a certified system is increased from 10 to 80% for most types of systems.
- A minimum allowable size reduction for certified transparent systems is instituted – reductions to transparencies resulting in less than 12 inches in length or width shall require further testing of the system.
- An important procedural aspect of this revision is the introduction of a procedure to retest systems at DS discretion in order to certify a system under this standard that was previously certified under Revision G or G (Amended).

Ballistic Testing

There are a number of changes to ballistic testing requirements incorporated into Revision H. The most significant change is associated with the increasing the number of test shots required for certification. Additionally, the procedure has changed and all shots of the same round are fired consecutively.

Forced Entry Testing

The most significant change to forced entry testing involves five-minute testing. Revision H institutes a new protocol which employs an identical toolset and six-man Test Team for 5, 15 and 60-minute forced entry testing. The toolkit has also been revised.

SPECIFIC CHANGES TO REVISION H

Changes for this version, presented in the approximate order in which they appear in this written standard:

1. Changed the standard's title, replacing "structural" with "building."
2. Changed the illustration on the front page.
3. Changed the level of signatory from DAS to C/PSP Deputy Assistant Director.
4. Changed the definition of "Bullet, Ball" and "Bullet Jacket."
5. Changed the definition of "certification letter" in order to reflect that a system must meet *all* of the requirements of this standard in order to be certified.
6. Removed the term / definition "certification of testing" (which was granted by a testing facility) in order to avoid potential confusion with certification issuance (which is granted by DOS).
7. Made the definition of "convoluted passage" broader.
8. Added a definition for "hand passageway."
9. Added a definition for "DOS Codes" and **Table 7: DOS Codes**.
10. Capitalized "DOS Model Number" throughout this Standard to reflect that it is a proper noun.
11. Removed the definition of "DS/PSD/SEP" as the branch no longer exists.
12. Replaced "FBO" with "OBO."
13. Added a definition for "FMJ."
14. Added a definition for "full-size systems" under the definition for "systems."
15. Added a definition for "manufacturer model number," clearly distinguished it from the DOS Model Number, and stated that both numbers should be unique.
16. Added a definition for "mounting gap."
17. Added a definition for "Technical Review." Capitalized "Technical Review" throughout this standard to reflect that it is a proper noun and a formal process.
18. Made it clear that "Test Personnel," "Test Team," and "Attack Team" are synonymous. Capitalized "Test Team" to indicate that it is a proper noun in this standard.
19. Removed the definition for "View Window" since the term does not appear elsewhere in this standard.
20. Stated test facilities must be approved by DS/PSP/PSD in order to perform SD-STD-01.01 testing and placed the definition for "independent test facilities" adjacent to the definition for "test facilities."
21. Stated the DOS is not obligated to issue ratings for products/systems it doesn't have a reasonable expectation of using.
22. Where appropriate, replaced "equipment" with "systems."
23. Created the new effective date / implementation date.

24. **Section 1.1: General** – Added verbiage that it is the test facility's responsibility to address safety and health practices and determine the applicability of regulatory limitations.
25. **Section 1.1.1: Users** – Clarifies that DOS as the certifying authority is not obligated to certify any system over which there is cause for concern.
26. Removes the ability to "grandfather" systems certified to a previous version of SD-STD-01.01; removes the term "recertification."
27. Removed ambiguity about modifications / changes to design – any modification(s) must go through the formal Technical Review process.
28. Added a procedure for systems previously certified under Revision G or G (Amended) of this standard to become certified under this version, Revision H. See **Section: 1.2.5: Previously Certified Systems**.
29. Certification Limits (new **Section 1.2.6**):

a. Moved it under **Section 1.2: Certification**, instead of under **Section 1.3: System Testing Concepts**.

b. Removed language that transparencies, louvers and grilles will not be limited to a certain size because it is not germane to this standard.

c. **Section 1.2.6(b): Certification for Sizes Other Than Test Sample Size** – Changed the maximum certified size for FE/BR systems, instituted a new minimum size reduction for transparencies, and clarified that aberrational shapes (e.g. a trapezoid) and aspect ratios require additional testing.

30. **Section 1.3.1(a): Ballistic Resistance Levels** and **Table 1: Ballistic Resistance Test Ammunition** and **Table 6: Minimum Ballistic Requirements** – Removed the submachine gun rating.
31. **Section 1.3.1(b): Shotgun** – Deleted "hand" from "passageways," and added that the Test Director can determine if shotgun testing is required.
32. Made it clear that multiple samples (regardless of the system's fielded size) may be needed for ballistic testing, and PSD's written permission is required for doing less than the number of shots in **Table 6: Minimum Ballistic Test Requirements**.
33. Added the **Multiple Samples** section. The manufacturer is allowed to submit multiple samples for testing. The manufacturer is allowed to submit up to one sample per round, or at the discretion of PSD, for ballistic testing. "Per round" does not mean "per shot" – switching out samples after every shot is not allowed. The manufacturer is allowed to submit up to one sample per vulnerable area for FE testing.
34. Removed the limitation on door sizes that had been Rev. G (Amended) **Section 1.3.4(b)**.
35. Where appropriate, specified that PSD approval must be required in writing.
36. **Section 1.3.5(b): Forced Entry Lock (FEL) Placement** – Stated FEL placement requirements may be waived for multi-point locking actuator.

37. Removed the old **Section 1.3.6** statement that separate samples may be submitted for testing of dissimilar portions because that is covered under the new **Section 1.3.4(c)**.
38. Changed the name of old **Section 1.3.7**, new **Section 1.3.8**, from "**1/4 Gap Certification**" to "**Mounting Gap**" (includes a definition for "mounting gap" in the **Definitions** section) – No longer requires 1/4″ as long as the FE/BR resistance of the gap is at least as strong as the system itself and the gap is the largest allowed by the manufacturer's installation instructions. Requires that the FE/BR performance of the mounting gap / mounting clearance to be tested.
39. **Section 1.4.1: Revocation Reasons**
    a. Specified that even one anomaly is sufficient cause for revocation of certification.
    b. Specified that even one material or design change is sufficient cause for certification revocation, and specified that the change did not have to be "significant."
    c. Specified that if a part of a system fails subsequent testing, that is sufficient cause to revoke the certification of a system.
    d. Specified that if a subsequent test failure causes concern about the body of knowledge upon which a certification is based, not just the test results upon which the certification was based, that is sufficient cause to revoke the certification of a system.
    e. Listed failure to comply with current OBO specifications as a reason for certification revocation; placed the burden of compliance on the manufacturer; and listed the appropriate remedy which is for the manufacturer to make the system compliant and submit the request via Technical Review.
    f. Listed cease of production (e.g. manufacturer goes out of business) as a reason for certification revocation.
    g. Listed failure to provide documentation as a reason for certification revocation.
40. **Section 1.4.3(e): Issue a Letter of Revocation** – Now, once a system's certification is revoked, that DOS Model Number cannot be used again, even if new testing and certification efforts are successful.
41. For procuring decertified systems, removed the "as a result of" clause, since it had no bearing on the rest of the section.
42. Under Revocation Procedures, changed "technical review" of the manufacturer's proposal to "engineering review." A "Technical Review" and/or retesting is still required if the manufacturer's proposal under **Section 1.4.3(b): Meet with Manufacturer and Review Manufacturer's Proposal** is inadequate.

43. Under QA/QC testing, replaced “materially different” with “different” because anomalies and imperfections are also covered under this clause.
44. Re-named **Section 1.6** as “**Point of Contact (POC) for this Standard**” instead of “**Availability**” and directed users to **Section 1.6: POC** elsewhere in this Standard instead of providing the contact information in multiple spots; also, updated the mailing address.
45. Reworded **Section 1.7.1: System Sample**.
46. Removed **Section 1.7.2: Independent Evaluation**.
47. Documentation section (**Section 1.8**) – Made it clear that all certification submissions (whether via testing or Technical Review) shall include the certification documentation specified in **Section 1.8: Certification Documentation**, and clearly required: title blocks, sheet numbers, to-scale drawings, revision blocks (if appropriate), Bill of Materials, notes (e.g. welding codes, tolerances, etc.), and clear labelling (e.g. attack side, protected side…) and the format shall be at DS/PSP/PSD’s discretion; specified that the metric units should be written first, with imperial units in parentheses.
48. Required that engineering drawings are submitted as one complete package.
49. Required a Certification Request Form for all requests for certification (whether via testing or Technical Review); previously, a letter was required.
50. Required the following in **Section 1.8.3: Installation Instructions**: care, maintenance, and repair instructions.
51. Witnessing – Changed the requirement from 10 to 15 days of advanced notice to be provided to DS/PSP/PSD in advance of the testing date.
52. **Section 1.9.2: Scheduling Procedure** – Changed the requirement from 10 to 30 days for the certification documentation to be submitted to the Test Director and DS/PSP/PSD prior to any scheduled testing.
53. Formatting – Separated out **Section 1.9.2** material into sub-sections for ease of reading – not a substantive change.
54. **Section 1.9.2(c)**: Explicitly required that the test sample comply with **Section 1.7.1: System Sample** before testing begins. Required DS/PSP/PSD to approve the testing schedule beforehand.
55. **Section 1.11: Testing and Certification of DOS-Designed FE/BR Systems** – Changed “DOS or A&E designed FE/BR equipment” to “DOS-designed FE/BR systems.”
56. **Section 1.12: Model Numbering Convention** – Tidied up the language with regard to assignment procedure, and removed the ability to assign interim model numbers. Created **Section 1.12.3: Unique Model Numbers Requirement** because unique model numbers are necessary for each system for identification and recordkeeping purposes. Made it clear that both DOS Model Numbers and manufacturer model numbers must be unique.
57. Deleted the old **Section 1.12.3: Prior Revision Certifications** (which concerned DOS Model Numbers) as it was no longer relevant under the new

Sections 1.2.2: Tested to Earlier Versions of SD-STD-01.01 and **1.2.5(a): New Certification of Previously Certified Systems**. Regarding DOS Model Numbers, for systems that were previously tested and are receiving new certification under this standard, see **Section 1.2.5(a)** as well as **1.12.1 DOS Model Number Assignment Procedure**.

58. **Section 2.1: Test Director** – Separated out the assigning of the Test Director from what the Test Director will ensure.
59. **Section 2.2** – Appended "**and Test Plan**" to the name of this section.
60. Reorganized the information in **Section 2.2: Test Sample Documentation (and Test Plan)** into **Section 2.2.1: Test Director Shall Examine Test Sample Documentation, and Formulate and Submit a Test Plan** and **Section 2.2.2: Test Sample Documentation Examination by Test Team**, although the content remains substantively the same except that language was added that DS/PSP/PSD's review will also "ensure that all potential vulnerabilities are investigated."
61. **Section 2.2: Test Sample Documentation and Test Plan** – Changed the DS/PSP/PSD advanced scheduling notice requirement from 10 to 15 working days.
62. **Section 2.3.2(b): Forced Entry Test Sample Installation** – Relocated language to here from the old **Section 2.5: Forced Entry Test Personnel** regarding the manufacturer's option to not allow active Test Team members to have participated in the installation.
63. **Section 2.3.2(d): Forced Entry Test Samples Which Require Footers** – Clarified that the bracing or capping is intended to represent the fielded condition and shall not enhance or detract from the FE performance – the intent of this section did not change.
64. **Section 2.3.3: Test Sample Labelling** – Required samples to be labelled with their attack and protected sides as well as their model numbers.
65. **Section 2.4.1: Rifle**

a. **Section 2.4.1(a): Test Sample Placement** – Made 20 feet the minimum placement.

b. **Section 2.4.1(b): Velocity Measurement Location** – Clarified what to do if measuring with other than screens.

c. **Section 2.4.1(c): Witness Panel Placement** – Provided a tolerance for the standoff of the witness panel.

66. Created **Section 2.4.3: Option to Waive Velocity Measurements** as a practical consideration, e.g. for oblique shots for which the measurements are difficult to take. Clarified that the option to waive any velocity measurements is at the sole discretion of DS/PSP/PSD representative.
67. **Section 2.5: Forced Entry Test Team** – Added requirements for the Test Team members to be "physically fit," "in good health," able to carry 60 lbs., and have a basic understanding of hand tool usage; lowered the minimum

weight to 150 lbs. from 160 lbs.; removed the upper weight limit which had been 250 lbs.; and increased the maximum age to 45 years old from 34 years old.

68. Reorganized the content of **Section 2.6**, formerly **Tools and Support Instrumentation**, and now **Ammunition, Tools, And Support Instrumentation**, including absorbing "old" **Section 2.7: Ammunition** – However, substantively there is no change. Moved witness panel specifics to new **Table 6** sub-table (not a substantive change).
69. **Section 2.6.1(a): Ballistic Testing Equipment** – Added new options besides screens for ballistic velocity measurement-taking.
70. New **Section 2.7: DS/PSP/PSD Representation** (old **Section 2.8**, same name) – directs the user to **Section 1.9.2(d): Witnessing and Scheduling** and **Section 2.2.1(c): Scheduling Testing** for the minimum required advanced notice and who provides this notice, to avoid double-listing the information in this standard under **Section 2.7**.
71. 2.7.3: Deleted section on "Non-Test Issues for DOS or A&E Designed System" because it is no directly applicable for inclusion into the testing standard.
72. **Section 3.1.2: Test Variations** – Added language that the DS/PSP/PSD representative may add to the test plan, although in practice this has always been true.
73. **Section 3.1.5: Door Operability** – This is outside the scope of this standard. Use OBO guidance.
74. **Section 3.2.1 Firing Procedure** – Formatting changed but there was no substantive change other than the change to the gap already mentioned in **Section 1.3.8: Mounting Gap**.
75. **Section 3.2.2(a)** – Removed the limit of six shots on transparencies – there is no upper limit.
76. **Section 3.2.3(a): Classified as "Complete Penetration"**
    a. Specified what purpose the light serves (non-substantive).
    b. Specified 800 lumens or brighter instead of a 60-watt lightbulb.
    c. Specified that penetration of the witness panel is a "Complete Penetration" and adjusted the **Definition** section's definition of "Ballistic Penetration" accordingly.
77. **Section 3.2.3(b): Classified as "No Penetration" or "Partial Penetration**" – Updated the title of this section and conformed to test house established test reporting conventions; updated the **Definitions** section accordingly.
78. **Section 3.2.5: Measuring Shot Velocity** (formerly part of **Section 3.2.4: Fair Hits**):
    a. Directed the user to **Section 2.4.3: Option to Waive Velocity Measurements**. Also, required that any instances of velocity measurements being waived shall be documented in the test report.

b. Referred the user to **Section 2.6.1(a): Ballistic Testing Equipment** instead of specifying that a chronograph must be used.

79. **Section 3.3.1(a): Interruptions** – Moved the verbiage on rests to its own section, new **Section 3.3.1(c): Rests**, though substantively there is no change. Also, since the old **Section 3.3.2(e)** mentioned periods of non-activity including "photography, safety inspection, conference with DS observers, etc.), these additional possible reasons for interruption were added/codified in the new **Section 3.3.1(a): Interruptions**.
80. New **Section 3.3.1(c): Rests** – As mentioned above, rest periods were moved out of **Section 3.3.1(a)** and into their own section. The language "rests may be given every 15 minutes" was replaced with "rests shall be given every 15 minutes" for 60-minute forced entry tests. Specified the duration (in minutes) that the rests shall be.
81. Old **Section 3.3.1(c)**, new **Section 3.3.1(d): Inspections** – Added verbiage that the Test Team may look or reach through the test sample; removed the verbiage about stepping behind the sample since it is already covered in new **Section 3.3.2(f): Stepping Behind the Test Sample**.
82. Moved Test Team selection information from **Section 3.3: Forced Entry Testing** to **Section 2.5: Forced Entry Test Team**, and clarified there that its the Test Director who determines who is on the Test Team.
83. **Section 3.3.1(e): Test Resources** – Clarified when tools shall not be used, because given that inspections are allowed during a test, tools are allowed during that time. Additionally, specified what condition the tools should be in – "serviceable" is not good enough. Clarified that the Test Team may reach through a sample to retrieve a tool if it is safe to do so. Moved **Table 2: Forced Entry Resources** note up to **Section 3.3.1(e)** that sharp edged tools shall not be re-sharpened during the test.
84. **Sections 3.3.2(c)** and **3.3.2(d)** were combined into new **Section 3.3.2(c): Exploit Dissimilar Portions** – The new section makes it clear that tests can be added to the test plan, not just subtracted, and includes a callback reference to **Section 2.2.1(b): Formal Test Plan**.
85. **Section 3.3.2(f): Stepping Behind the Test Sample** – This section serves the same purpose as the old **Section 3.3.2(g)**.
86. Old **Section 3.3.2(h)** – A prohibition on the use of the wood ax during five-minute forced entry testing, was removed.
87. Old **Section 3.3.2(i)** – A prohibition on the use of the wood-splitting maul, was removed.
88. Section 3.4.1 Ballistic Rejection Criteria – Clarified language to conform with complete penetration, partial penetration and no penetration definitions.
89. **Section 3.4.2(a): Pass/Fail Shapes** – Clarified that it is the exterior (outer-most) dimensions provided for the pass/fail shapes.

90. **Section 3.4.2(b): Inherent Shape Advantage for Forced Entry** – Provided guidance for systems that are inherently not vulnerable to the pass/fail shape sizes in **Section 3.4.2(a)**.
91. Removed old **Section 3.4.3: Door Operability Rejection Criteria**, because having a pull/push force exceeding 12 lbs. (5.5 kg.) is no longer a reason for a door to fail FE/BR testing.
92. **Section 3.7: Safety** – Provided a short list of possible considerations, keeping the language that the Test Director is responsible for safety; the Department of State is still not responsible for safety; stated that this standard does not address all safety concerns.
93. **Section 4.1: General** – Specified that the test lab supplies the test reports to DOS, not the manufacturer, thus clearing up chain of custody issues for the reports; changed the language concerning proprietary information and consent to documentation release.
94. **Section 4.2.3(b): Verification of Identical System** – This section requires verification by the test facility that the test sample was identical to the system shown in the manufacturer's submitted drawings.
95. **Section 4.2.5: Test House Test Report** – Required that the test report include the as-tested size of the mounting gap.
96. **Section 4.2.6: Ballistic Test Data** – Took out requirement to weigh bullets.
97. **Section 4.2.8: Photographic Record of Testing** – Required images to be in color, in focus, high-res, and suitable for printing.
98. **Section 4.2.10: Video Recording of Testing** – Removed an archaic video format, VHS; stated that FE video footage should be recorded from both the attack side and the protected side, made it clear video was only for the FE portion of the testing, and included a requirement for a watermark.
99. **Section 4.3: Certification Issuance:**
    a. Deleted language indicating that the testing laboratory will provide DS/PSP/PSD with test reports for each system tested because it was duplicative of the language in **Section 4.1: General Reporting**.
    b. Added the requirement for OBO guidance to be met in order for certification issuance to occur.
100. **Section 5.1: Certification by Technical Review** – Clarified that *all* T/Rs (not just those for upgraded systems) are done only at DS/PSP/PSD's discretion. Moved "DOS reserves the right to grant or reject a request for certification" from **Section 5.3: Technical Review Procedure** to **Section 5.1**. Added that any test reports for previously tested systems should be included with the request. Stated that new testing and/or retesting may be required (see also **Section 5.3.3: New Testing and/or Retesting**). Changed "previously certified system" to "each previously certified and/or tested system" because under the limits of **Section 5.1** you may receive a T/R for an upgraded non-certified system.

101. **Section 5.1.1: Change to Certified Systems** – Clarified that a T/R is needed even if the change is done merely to comply with OBO specifications.

102. Added **Section 5.1.4: Request for Certification Under Current Version of this Standard**

103. **Section 5.2: Requests for Certification by Technical Review** – Made it clear that these are only the minimum requirements. Required an explanation of each difference between prior systems and the candidate system. Spelled out which documents from **Section 1.8: Certification Documentation** are required for a T/R.

104. **Section 5.2.3: Requests For Certification By Technical Review** – Clarified what must be included in the request.

105. **Section 5.3: Technical Review Procedure** – Clarified that the candidate system will only be compared with its manufacturer's other systems.

106. **Table 1: Ballistic Resistance Test Ammunition** – Required M80 to be steel jacketed; copper-jacketed M80 rounds are no longer acceptable for testing to this standard.

107. **Table 2: Forced Entry Test Resources**:
   a. Added a number of replacements for each resource.
   b. Removed the restriction that the wood-splitting maul shall not be used when testing transparent portions of samples.
   c. Replaced the 12-lb. sledgehammer with the 10-lb. sledgehammer.
   d. Added 14″-handle hatchet.
   e. Added 17″-handle medium axe.
   f. Removed 3.5-lb. 3″-head wood axe.
   g. Added 5-lb. 36″-handle pick mattock.
   h. Added 10″ compact grip 24 TPI mini hacksaw and replacement blades.
   i. Added 12″ wood hand saw.
   j. Replaced 48″ bolt cutters with 42″ bolt cutters.
   k. Added 24″ pipewrench.
   l. Removed portable propane torch.
   m. Added 2x4x72″ lumber.
   n. Added wood cribbing, both 6x6x16″ and 4x4x16″.
   o. Added round 2x60″ schedule 40 steel pipe.
   p. Added square 2x2x72″ steel tube.
   q. Added socket sets, both SAE and metric.
   r. Added hex key sets, both SAE and metric.
   s. Added metal retractable utility knife and replacement blades.
   t. Added one knife, similar to the LPS1175 Tool Category A knife.
   u. Added Nylon 3-strand 0.5″ diameter 29′-length rope, similar to the LPS 1175 Tool Category A rope.
   v. Added metal 8″ hand file.
   w. Added 15- and 60-minute tools to 5-minute test.

108. **Table 6: Minimum Ballistic Requirements**:
  a. Removed the submachine gun.
  b. Increased the number of shots per round for Opaque Areas: Center of Area from 1 to 8.
  c. Increased the number of shots per round for the M855 for Opaque Areas: Seams, Framing, Etc.
  d. Increased the number of shots per round for Transparencies: Center of Area from 1 to 8.
  e. Created a “Witness Panel” sub-table (not a substantive change).
  f. Narrowed the range of acceptable ‘e’ (thinner) witness panel aluminum foil, and provided thickness ranges, hardness, and yield strength for both witness panels.
  g. Referenced a new figure, **Figure 3: 8-Shot Spacing**, vice describing the spacing in the Notes text.
  h. Created and referenced a new figure, **Figure 4: 4-Shot Spacing**.

109. **Table 7: DOS Codes** – Simplified the first digit to be either 1 or 2 (transparent or opaque).

110. Created **Table 8: Test Labs Approved by DOS to Conduct FE Tests** to this Standard.

111. Removed the form, **Certification of Testing**, for the following reasons: 1) “Certification” is only approved by DS/PSP and 2) the door pull test part of the form may not conform to current OBO guidance.

The U.S. Department of State program:

SD-STD-01.01 is part of the DOS program to ensure forced entry and ballistic resistant equipment which meet DOS standards are placed in DOS buildings overseas. This program accomplishes:

- Establishing standards for forced entry and ballistic resistant levels of protection as a function of potential threat levels.
- Establishing test criteria to ensure equipment meets the forced entry and ballistic resistant needs of the DOS.
- Ensuring systems also meet the U.S. Department of State Bureau of Overseas Buildings Operations (OBO) *Specifications for Forced Entry/Ballistic Resistant Doors, Windows, Modular Panels, Glazings and Louvers*.

Requests for copies, questions, concerns, or comments concerning this Standard may be directed to the contact in **Section 1.6: POC**.

DEFINITIONS

A&E – Architectural and Engineering.

Assembly Drawing – A drawing which shows how a system is assembled. It includes the locations and arrangements of all bolts, glazing stops, glazing spacers, etc.

Ballistic Penetration

- Complete Penetration – A perforation of the witness panel through which light passes (see **Section 3.2.3(a): Classified as "Complete Penetration"**).
- No Penetration – No deformation of strike-face of test sample.
- Partial Penetration – A penetration of the test sample that is not classified as "Complete Penetration" or "No Penetration."

BR – Ballistic Resistant.

Bullet, AP – An armor-piercing bullet which contains a core of hardened material to penetrate armor and other materials which may resist penetration by lead-cored bullets.

Bullet, Ball – Military designation of a small-arms projectile (bullet) consisting of a soft core (often lead) encased in a shell (jacket) of harder metal.

Bullet Jacket – The thin shell of material which encases the core of a bullet. The jacket material usually is copper or steel. A copper wash may be used to prevent rust.

Bullet Yaw – The angle between the centerline of a bullet and its flight path.

Candidate System – A manufacturer or DOS-designed system which is being considered by DS/PSP/PSD for forced entry and/or ballistic resistant certification.

Certification Letter – Document issued by DS/PSP/PSD for systems which meet the certification requirements as stated in this standard.

Contracting Officer (CO) – A person with the authority to enter into, modify, administer, and/or terminate contracts and make related determinations and finding.

Contracting Officer's Representative (COR) – A person who is designated by the Contracting Officer to assist in administering the technical aspects of the contract.

Convoluted Passage – Any passageway through a system. Can be direct, indirect, or intricate.

Deal Tray – The subassembly of a transaction window which enables routine passage of materials of limited size between sides of the window. The window may be fixed, have a convoluted design, or include moveable elements to enhance security.

Disturbed Area – The area surrounding the point of ballistic impact which has been visibly cracked, holed, distorted, or otherwise changed by the ballistic impact.

Door, Double – Assembly of two doors with an opening typically twice as wide as either single door with a common latch and lock edge.

Door Threshold – The sill of a doorway.

DOS – U.S. Department of State. When "Department" is written in the context of this standard, it refers to the U.S. Department of State.

DOS Code – A four-digit code as detailed in **Table 7: DOS Codes** – this shorthand format helps end users quickly specify their rating needs – a system's DOS Code is separate from and not to be confused with its DOS Model Number.

DS – Bureau of Diplomatic Security, U.S. Department of State.

- DS/PSP/PSD – Physical Security Division, Office of Physical Security Programs, Bureau of Diplomatic Security.

Escape Hatch – A horizontally or vertically mounted non-door assembly. It may be roof-mounted and include a collapsible interior ladder.

FE – Forced Entry.

- FE/BR – Forced Entry/Ballistic Resistant.

FMJ – Full Metal Jacket.

Hand Passageway – This includes but is not limited to the passageways inherent in deal trays, transaction drawers, louvers, and gun ports.

Hardline – Partitioning structure or system providing 5, 15, or 60-minute forced entry and/or ballistic resistant protection in order to preclude unauthorized access to a core or protected area.

Impact Obliquity – The angle between the bullet path and a line which is perpendicular to the impact surface at the point of impact.

Implementation Date – Date after which this standard supersedes all previous revisions.

Letter of Revocation – Notification to a manufacturer by DOS that a previously certified FE and/or BR system has been decertified.

Letter of Suspension – Notification to a manufacturer by DOS of its intent to decertify a previously certified FE and/or BR system.

Louvers – Angled and gapped slats which permit the passage of air through an otherwise impassable vertical barrier. While louvers usually are rigidly constructed, the slats may be moveable to seal off air passage.

Model Number – A unique identifier for a manufactured system.

- DOS Model Number – The unique identifier for every FE/BR system certified by the U.S. State Department. This model number is assigned by the U.S. State Department. This model number will follow the conventions of **Figure 1: DOS Model Numbering Convention**.
- Manufacturer Model Number – A unique identifier for a manufactured system which is assigned by the system's manufacturer.

Mounting Gap – "Mounting gap" is synonymous with "installation mounting gap." In order for a system, such as a door or window, to be installed into a building structure, a rough opening is needed. The gap between a system and the rough opening into which it is installed is called the mounting gap.

Mulled – The physical connecting together of two systems. The two systems may be anchored directly to each other or have a mullion between them.

- Mullion – A component which separates two adjacent and separate systems. It can be vertical or horizontal, movable or fixed. For purposes of this standard, a mullion does not include steel or concrete structural members which are present in the building. Seismic joints are to be considered as mulled connections.

OBO – U.S. Department of State Overseas Building Operations, formerly the Office of Foreign Building Operations (FBO).

SD-STD-01.01 – State Department Standard 01.01, Certification Standard, Forced Entry and Ballistic Resistance of *Building* Systems, formerly Certification Standard, Forced Entry *and Ballistic* Resistance of Structural Systems, and before that, formerly Certification Standard, Forced Entry Resistance of Structural Systems).

SD-STD-01.02 – Former State Department Standard 01.02, Certification Standard, Ballistic Resistance of Structural Systems, superseded by SD-STD-01.01, Revision E.

Spall – Bullet or target material which has broken free due to ballistic impact and projected into the surrounding area.

System – The assembly of structural elements and devices which compromise the forced entry and/or ballistic barrier used by the U.S. Department of State. These assemblies can be designed and produced by manufacturers or designed by the Department of State for construction by contractors.

- DOS-Designed Systems – These include concrete and steel hardlines, seismic joints, and specialty items, such as doors, windows, grilles, etc. for security upgrade projects.
- Full-Size Systems – Systems that are representative of the fielded system's actual size.
- Manufacturer-Delivered Systems – These include both individual and mulled systems.
    - Individual Systems – Individual doors, windows, louvers, hatches, grilles, or wall panels. The doors, windows, and panels may have two or more transparencies separated with a mullion.
    - Mulled Systems – Individual systems anchored together to make a larger system.

Technical Review – A Technical Review, abbreviated in writing by "T/R," is the review process described in **Section 5.0: Technical Review** by which a system may be granted certification at the discretion of DS/PSP/PSD if the system is sufficiently similar to one or more previously tested systems. See **Section 5.1: Certification by Technical Review (T/R)** for limitations. Note that new testing and/or retesting of a candidate system may be required under **Section 5.3.3: New Testing and/or Retesting**.

Test Director – Individual identified by the test facility responsible for the overall conduct, data collection, and safety of a forced entry or ballistic resistant test.

Test Facility / Test Lab – Laboratory or other area where forced entry and/or ballistic resistant testing is conducted. It may be either government-operated or independent. The facility must be approved by DS/PSP/PSD in order to perform SD-STD-01.01 certification testing; DS/PSP/PSD maintains a list of the facilities approved to perform this work.

- Government Test Facility – Test facility operated by the U.S. Government. Can be used for DOS-designed systems or for QA/QC purposes at the discretion of DOS.
- Independent Test Facility – Test facility which is independently owned and operated. Not affiliated with any other corporation, interest group, or government agency.

Test Fixture – The structural assembly which holds the test sample. Field mounting hardware is used to mount the test sample into the test fixture.

Test Sample – The forced entry and/or ballistic resistant system(s) which is/are to be tested to this standard.

- As-Tested Size – The size of the system's full-scale test sample.

Test Team – When referring to forced entry testing, the Test Team is the team of personnel actively engaged in the Concentrated Assault Test but not including the Test Director, data recorders, or other supervisory personnel. Synonymous with “Test Personnel” and “Attack Team.”

Transparent Element – A glazing panel, including those designed to be backpainted, coated, or filmed, and including those which have been made only semi-transparent, or even opaque. Synonymous with “transparency” and “transparent armor.”

Window Frame – The opaque portion of a transparent assembly into which the transparent element is mounted.

Window Grille – Spaced, rigid bars which are mounted over exterior windows of the host structure to prevent access.

Witness Panel – A panel of thin material that is positioned behind a test sample during ballistic testing, the perforation of which determines penetration regardless of the penetrating material.

1.0 SCOPE AND REQUIREMENTS

1.1 GENERAL

This standard supersedes all prior versions of SD-STD-01.01 and SD-STD-01.02. It sets forth the requirements and testing procedures to certify forced entry (FE) and/or ballistic resistant (BR) systems which are intended for use by the U.S. Department of State (DOS) in its diplomatic facilities throughout the world. The certification of FE and BR systems is mandatory and indicates that the systems are rated to provide the level of FE and BR protection required by DOS standards for specific facilities in certain threat environments. This standard does not purport to address all of the safety, environmental, and end-user concerns, if any, associated with its use. It is the responsibility of any testing facility that conducts testing to this standard to establish appropriate safety and health practices, and determine the applicability of regulatory limitations, before conducting any such testing (see also **Section 3.7: Safety**).

1.1.1 Users

1.1.1.a Intent

This standard is intended primarily for manufacturers to test their products/systems and for DOS to certify systems that meet DOS needs.

1.1.1.b Use

DOS is not under any obligation to issue FE/BR ratings and/or certifications for products/systems. For example, if products/systems are of a type that DOS does not have a reasonable expectation of using, then DOS is not obligated to certify them. If any reason comes to light before, during, or after testing which DS feels is cause for concern about a system, then DOS is not under any obligation to issue FE/BR ratings and/or certifications for that system, regardless of testing outcomes.

1.1.1.c DOS-Designed Systems

This standard also can be used to test and certify DOS-designed walls, seismic joints, doors, grilles, etc. **Section 1.11: Testing and Certification of DOS-Designed FE/BR Systems** details considerations for testing DOS-designed FE/BR systems.

1.1.2 Exceptions

This standard does not apply to items of personal protection (helmets, visors, vests, etc.) or to materials intended for installation in automobiles or other mobile structures.

1.1.3 Implementation

1.1.3.a Implementation Date

This standard will take effect as of the date it was signed by the DS/PSP Office Director (see signatory page). All FE/BR products procured pursuant to DOS physical security standards shall require certification to SD-STD-01.01.

1.1.3.b List of FE/BR-Certified Systems

The Department of State Bureau of Diplomatic Security shall maintain both a list of FE/BR systems certified under Revision G or G (Amended) of SD-STD-01.01 hereafter referred to as the “G List,” and a list of FE/BR systems certified under this revision of SD-STD-01.01, Revision H, hereafter referred to as the “H List.” A system’s DOS Model Number will differentiate to which list an FE/BR system belongs under **Section 1.12.1: DOS Model Number Assignment Procedure**. Should the manufacturer of a system on the G List desire the system to be placed on the H List, the manufacturer should follow the guidance of **Section 1.2.5: Previously Certified Systems** (see also **Section 1.2.2: Tested to Earlier Versions of DS-STD-01.01**).

1.1.3.c Procurement Relative to the Implementation Date

Products on the G List may be procured if the procurement action was initiated two years or less after the implementation date this version of SD-STD-01.01, Revision H.

1.2 CERTIFICATION

Certifications may be granted to systems which fall under **Section 1.2.1: Tested to Current Version of SD-STD-01.01** or **Section 1.2.2: Tested to Earlier Versions of SD-STD.01.01**.

1.2.1 Tested to Current Version of SD-STD.01.01

Candidate systems which successfully complete the testing requirements specified in this version of SD-STD-01.01 may be granted certification.

1.2.2 Tested to Earlier Versions of SD-STD.01.01

Any system tested to a version of SD-STD-01.01 earlier than Revision G shall not be certified under Revision H without going through the entire certification process anew, as though the system had never been tested or certified before.

Candidate systems which were previously certified to Revision G or Revision G (Amended) of SD-STD-01.01 and meet the requirements of **Section 1.2.5(a): New Certification of Previously Certified Systems** may be granted certification under Revision H upon successful Technical Review (see **Section 5.0: Technical Review)**.

1.2.3 Certification Submission

Certification submittals for candidate systems must comply with the documentation requirements as specified in **Section 1.8: Certification Documentation**, and **Section 5.0: Technical Review** for Technical Reviews. Certification submittals for Technical Review which do not require new or retesting need not include a Test-To-Failure Agreement.

1.2.4 Modifications

Once a system is certified to this Standard, no design change or change in component material shall be made to that system. Items that are not relevant to the FE/BR certification are not subject to this design change requirement, for example: coatings or tints to glazings (except if they are applied films or interlayers that change the physical construction), paints/anodizing, DOS-approved hardware schedules, and cladding or decorative trim pieces that were not part of the test sample.

For a new system that is substantially similar to a certified system, the new system may be approved via the Technical Review process in **Section 5.0: Technical Review**. If the Technical Review is approved by DOS, then the new system must have a DOS Model Number change and a manufacturer model number change to reflect this modification (see **Section 1.12: DOS Model Numbering Convention**).

1.2.5 Previously Certified Systems

1.2.5.a New Certification of Previously Certified Systems

Candidate systems which were previously certified to Revision G or Revision G (Amended) of SD-STD-01.01 may apply for a new certification under Revision H at the discretion of the U.S. Department of State. The new certification will require a Technical Review and may require retesting and/or new testing in order to fully evaluate the system (see **Section 5.0: Technical Review**). The onus is on the manufacturer to apply for a new certification, including submitting a letter of intent to become certified under Revision H and a complete set of drawings. If the request for new certification is approved by DS/PSP/PSD, then a new certification letter will be issued, and a new DOS Model Number will be issued (see also **Section 1.12: DOS Model Numbering Convention**).

1.2.6 Certification Limits

1.2.6.a Orientation

Certification shall be granted only for orientations that are ± 30º (about a horizontal axis) from the test sample's installation orientation. For example, a window that was tested perpendicular to grade cannot be installed parallel to grade and be called a skylight.

1.2.6.b Certification for Sizes Other Than Test Sample Size

The as-tested size is the size of the system's full-scale test sample.

1.2.6.b.1 Sizes Which Exceed Eighty Percent in Any Dimension

Certification will not apply to sizes which exceed the as-tested size by more than 80% in any dimension. Sizes larger than 80% in any dimension may be certified in accordance with **Section 5.0: Technical Review** at DS/PSP/PSD discretion. This may require additional testing. If certification is granted, then the new size will be assigned a new DOS Model Number.

1.2.6.b.2 Glazing Openings Smaller than Twelve Inches

A window (or vision) system with glazing openings smaller than 12 inches in length or width must be tested at those dimensions in order to be certified. Certification of a system with a larger as-tested size will not extend to sizes with glazing openings smaller than 12 inches in length or width.

1.2.6.b.3 Aberrational Shapes

A system is certified for its as-tested shape, e.g. a four-sided shape with 90-degree corner angles. Any aberration from the as-tested shape is required to undergo additional testing to determine its FE/BR resistance, and will receive a new DOS Model Number if it is successful in providing the required level of protection (see also **Section 1.12: DOS Model Numbering Convention**). For example, if the as-tested shape was a rectangle, then more testing is required for any of the following shapes: diamond or rhombus, trapezoid, circle or semicircle, hexagon, and octagon.

1.2.6.b.4 Aberrational Aspect Ratios

The certification applies only to aspect ratios that are similar to the as-tested size (height / width). Sizes with dissimilar aspect ratios to the as-tested size may be certified in accordance with **Section 5.0: Technical Review** at DS/PSP/PSD discretion. This may require additional testing. If certification is granted, then a new DOS Model number will be assigned (see also **Section 1.12: DOS Model Numbering Convention**).

1.3 SYSTEM TESTING CONCEPTS

1.3.1 Ballistic Resistance (BR)

1.3.1.a Ballistic Resistance Levels

U.S. Department of State policy establishes the following levels of small arms ballistic protection (see also **Table 1: Ballistic Resistance Test Ammunition** and **Table 6: Minimum Ballistic Requirements**):

- N – No ballistic resistance
  - For systems which have not been ballistic-tested, or which have failed ballistic testing to this standard
- R – Protection from military rifle projectiles
  - For systems which successfully pass rifle ballistic testing per this standard
- AP – Protection from Armor Piercing (AP) projectiles
  - This testing is optional and typically reserved for special projects – The number of shots and witness panel will be determined on a case-specific basis by DS/PSP/PSD
- SH – Shotgun (see **Section 1.3.2(b): Shotgun**)

1.3.1.b Shotgun

The shotgun (see **Section 2.4.2: Shotgun** and **Table 6: Minimum Ballistic Test Requirements**) is used to test the ballistic vulnerability of any passageways which a candidate system may possess. DS/PSP/PSD and/or Test Director shall determine if shotgun testing is required.

1.3.2 Forced Entry (FE) Resistance

1.3.2.a Forced Entry Resistance Levels

U.S. Department of State policy establishes the following levels of forced entry protection:

- None
- 5 minutes
  - For systems which pass a minimum of five (5) minutes of concentrated assault per this standard for each dissimilar area
- 15 minutes
  - For systems which pass a minimum of fifteen (15) minutes of concentrated assault per this standard for each dissimilar area
- 60 minutes
  - For systems which pass a minimum of sixty (60) minutes of concentrated assault per this standard for each dissimilar area

1.3.3 Concurrent Testing

This standard may be used to evaluate the resistance of a system to either a ballistic threat or a forced entry threat. Concurrent testing evaluates the resistance of the system to both types of threats. DS/PSP/PSD shall determine if concurrent testing is required.

1.3.4 Multiple Samples

1.3.4.a Damage from a Previous Test

No test of a dissimilar area shall be waived due to damage from a previous test. All dissimilar areas shall be evaluated.

1.3.4.b Required to Provided Multiple Samples

A manufacturer may be required by DS/PSP/PSD and/or the test facility to provide multiple samples for ballistic testing in order to fully evaluate the system.

1.3.4.b.1 Testing Less than the Minimum Required Number of Shots

Testing less than the number of shots required by **Table 6: Minimum Ballistic Test Requirements** is not permitted without DS/PSP/PSD's written permission. This written permission is required regardless of the candidate system's as-tested size, and should state in detail how many shots DS/PSP/PSD will require in order to evaluate this system. For example, if the test sample's size presents an inherent shape advantage for ballistic testing such that not all of the shots required by **Table 6: Minimum Ballistic Test Requirements** will fit on a single sample and DS/PSP/PSD written permission is not obtained, then the manufacturer shall be required by DS/PSP/PSD and/or the test facility to provide multiple test samples sufficient to complete all of the shots required by **Table 6**. The number of shots provides statistical validity of the system's ability to resist ballistic threats, not just a demonstration of what a single fielded system may experience under adverse conditions.

1.3.4.c Electing to Provide Multiple Samples

1.3.4.c.1 Electing to Provide Multiple Samples for Ballistic Resistance

The manufacturer may elect to provide multiple samples for ballistic testing, up to one per round (not to be confused with being allowed to provide one sample per shot), or at the discretion of DS/PSP/PSD, depending on the system's type and size.

1.3.4.c.2 Electing to Provide Multiple Samples for Forced Entry

The manufacturer may elect to use a different sample for forced entry testing than for ballistic testing, except if **Section 3.4.2(b): Inherent Shape Advantage for Forced Entry** is applicable. The manufacturer may elect to provide multiple samples for forced entry testing, up to one sample per dissimilar area.

1.3.5 Doors

1.3.5.a Door System Undercuts

1.3.5.a.1 Doors with Thresholds

If the door has a threshold (i.e. steel threshold bar), the undercut between the bottom edge of the door and the top of the threshold, shall not exceed 1/4″ (6 mm). The total distance from the bottom edge of the door to the top of the slab shall not exceed 3/4″ (19 mm).

1.3.5.a.2 Doors without Thresholds

If the door has no threshold, the undercut shall not exceed 1/4″ (6 mm).

1.3.5.b Forced Entry Lock (FEL) Placement

No door system will be certified which locates a Forced Entry Lock (FEL) actuating device (e.g., door handle, thumb turn, or keyway) less than 10 inches (250 mm) from the bottom of the door or which locates a forced entry lock actuating device more than 30 inches (760 mm) above the passage lock set unless specific DS/PSP/PSD approval in writing is obtained. This requirement may not apply to doors which employ a multi-point locking system that is retracted from a single actuator.

1.3.6 Mulled Systems

1.3.6.a Mulling for Testing

The manufacturer may mull individual systems together into the test frame and can request certification for the individual systems, mullions, and mulled joints.

1.3.6.b Schematic

If this option is elected, the manufacturer shall provide a schematic of the proposed test configuration as part of the drawings submitted in **Section 1.8.2: Engineering Drawings and Specifications**.

1.3.7 Test Complete Systems

Only complete systems shall be tested. The deal tray in a teller window, for example, cannot be tested separately from a window without specific DS/PSP/PSD approval in writing.

1.3.8 Mounting Gap

It is up to the manufacturer to determine the installation mounting gap required for their candidate system.

It is the manufacturer's responsibility to ensure that their mounting gap does not exceed OBO guidelines and project-specific details.

The test sample shall be fabricated, assembled, installed, and tested with the maximum allowable mounting gap specified in the manufacturer's installation instructions (see also **Section 2.3: Test Sample Mounting and Section 2.3.2(c): Forced Entry Mounting Clearance**).

The standard manufacturer's mounting gap shall not exceed the tested size as specified in the test report (see **Section 4.2.5: Test House Test Report**).

Certification to this standard requires that the FE and/or BR resistance of the mounting gap is at least as strong as that of the system itself.

The FE and/or BR resistance of the mounting gap shall be verified by FE and/or BR testing (see also **Section 2.3: Test Sample Mounting**, **3.2: Ballistic Testing**, and **3.3.5: Mounting Gap Forced Entry Testing**).

1.4 REVOCATION OF CERTIFICATION

1.4.1 Revocation Reasons

Certification may be revoked for cause at the discretion of DOS for any of the following reasons.

1.4.1.a Anomaly

There is one (or more) minor anomaly or imperfection in a manufactured system.

1.4.1.b Material or Design Modification

There is one (or more) material or design modification to a certified system which reduces DOS confidence in the performance characteristics of that system.

1.4.1.c Component Substitution

Other than as allowed under **Section 1.2.4: Modifications**, one (or more) component(s) (e.g., hardware, locking devices, etc.) is substituted in a finished product.

1.4.1.d QA/QC Failure

The system fails to pass any DOS Quality Assurance/Quality Control (QA/QC) evaluation or testing (see **Section 1.4.4: Quality Assurance/Quality Control (QA/QC)**).

1.4.1.e Subsequent Test Failure

The system or a part thereof fails to pass any subsequent test, causing sufficient concern about the body of knowledge upon which the system's certification was based.

1.4.1.f Failure to Comply with OBO Specifications

OBO specifications evolve over time. It is the manufacturer's responsibility to ensure that the system continues to meet OBO specifications. If the system fails to comply with current

OBO specifications, then the manufacturer shall update their system in order to make it comply and submit a Technical Review per **Section 5.0: Technical Review**.

1.4.1.g Failure to Provide Requested Documentation or Lack of Documentation

The manufacturer shall provide a copy of complete drawings of the system and the system's certification letter upon request by DS/PSP/PSD. Failure to provide such documentation to DS/PSP/PSD in a timely manner is sufficient cause for DS/PSP/PSD to revoke certification until such documentation is provided.

The manufacturer shall provide installation instructions that comply with OBO requirements upon request by OBO or DS/PSP/PSD. For a description of what constitutes acceptable installation instructions, see **Section 1.8.3: Installation Instructions**.

1.4.1.h Production Ceased

If the original manufacturer ceases to produce the system, then the system will no longer be certified. This may happen, for example, if the manufacturer is bought out, closes, or switches target markets. If another manufacturer wishes to produce the previously certified system, FE/BR certification does not automatically convey with the designs. The new manufacturer must submit a written request for the system to keep its status as a certified system under the new manufacturer's name and with a new DOS Model Number (see **Section 1.12: Model Numbering Convention**).

1.4.2 Procurement of Decertified Systems

A system whose certification status changes from certified to uncertified may be procured by DOS or its contractor, at their discretion, if the system's certification was valid when the procurement action was initiated.

1.4.3 Revocation Procedures

Should there be cause to revoke the certification of a DOS-certified system, DOS will take the actions in **Section 1.4.3: Revocation Procedures**.

1.4.3.a Notify Manufacturer

DOS will notify the manufacturer that it has **30 calendar days** to meet with DOS to discuss its plans to redress the shortcomings in question. The manufacturer should come prepared to discuss the deficiencies with appropriate drawings and specifications.

1.4.3.b Meet with Manufacturer and Review Manufacturer's Proposal

DS/PSP/PSD will meet with the manufacturer. DOS will review the manufacturer's proposal to correct the shortcomings, e.g. correcting system deficiencies by material change or design modification, and DOS will make a determination within **10 working days**, based upon an engineering review, as to the adequacy of the proposed improvement/remedy.

If the manufacturer's proposal is determined by DOS to be adequate, then DOS will so notify the manufacturer and no further action will be taken, provided that the manufacturer provides the complete up-to date drawings. The up-to-date drawings shall depict all proposed fixes.

1.4.3.c Issue a Letter of Suspension

If the manufacturer's proposal is determined by DOS to be inadequate, then DOS will then issue a Letter of Suspension notifying the manufacturer of DOS' intent to decertify the system(s) in question.

1.4.3.d Grace Period

The manufacturer will be granted a period of **60 calendar days** to correct deficiencies by material change or design modification. A Technical Review which may require new or retesting (at the manufacturer's expense and at the discretion of DOS) will be required to obtain new certification and, if the Technical Review is successful, a new DOS Model Number will be granted. Extensions to the grace period are at the discretion of DOS and must be obtained from DS/PSP/PSD in advance and in writing.

1.4.3.e Issue a Letter of Revocation

DOS will issue a Letter of Revocation at the conclusion of the grace period if identified deficiencies have not been corrected to the satisfaction of DOS. Once a Letter of Revocation has been issued, subject systems must be retested, at the manufacturer's expense, in order to be eligible for certification. If retesting and new certification efforts are successful, then a new DOS Model Number will be issued.

1.4.4 Quality Assurance/Quality Control (QA/QC)

1.4.4.a QA/QC Procurement

DOS reserves the right to implement quality assurance/control (QA/QC) inspections and/or testing. This may involve the procurement of certified systems from the manufacturer at the expense of the DOS.

1.4.4.b Differences

Certified systems procured by DOS for QA/QC or other purposes which are different from the approved design will be grounds for initiating revocation of certification procedures under **Section 1.4.3: Revocation Procedures** (see also **Section 1.4.1: Revocation Reasons**).

1.4.4.c Test Failure

Certified systems tested by DOS for QA/QC or other purposes which fail independent testing, for whatever reason, will be grounds initiating revocation of certification procedures under **Section 1.4.3: Revocation Procedures** (see also **Section 1.4.1: Revocation Reasons**).

1.5 REVISIONS

Revisions of any portion of this standard must be approved by DS/PSP/PSD. All requests for revisions will be forwarded to the addressee listed in **Section 1.6: Point of Contact for this Standard**.

1.6 POINT OF CONTACT (POC) FOR THIS STANDARD

Additional copies of this standard may be obtained from, and questions, comments, or suggestions should be directed to:

- U.S. Department of State, Attn: DS/PSD/RD, SA-14 Building, 9th Floor, Washington, D.C., 20520-1403
- dspsprd@state.gov

1.7 TEST SAMPLE

1.7.1 System Sample

Systems submitted for testing must:

- Comply with **Section 1.3.7: Test Complete Systems**.
- Be full-size systems, representative of the fielded system's actual size.
- Include all required anchor bolt system hardware, using the maximum anticipated bolt spacing and the minimum anticipated bolt sizes and strengths.
- If it is a door system, include all Forced Entry Locks (FELs), a functional approved passage set, and the standard approved door closer.
- If the system moves or is operational (e.g. a door or hatch), it shall at minimum include all devices required for operation.
- Be in "ready to purchase" condition.
- Be able to meet all current applicable OBO specifications.

1.8 CERTIFICATION DOCUMENTATION

All submissions for certification shall include the following documentation. The documentation format shall be determined by DS/PSP/PSD.

1.8.1 Request for Certification

Whether requesting certification via testing or via Technical Review (see **Section 5.0: Technical Review**), the manufacturer shall submit a formal **Certification Request Form** to the email address listed in **Section 1.6: Point of Contact for this Standard** specifying the type of system (e.g. door, window, etc.), the protection rating(s) sought, and, if applicable, what tests are to be performed.

1.8.2 Minimum Requirements for Engineering Drawings and Specifications

This section provides the minimum requirements, all of which must be met. Drawings shall be submitted as **one** complete package (vice multiple files).

Note: All proprietary information submitted will be treated as company confidential and will not be disclosed to unauthorized personnel. Manufacturer shall mark all documentation as company confidential as appropriate.

1.8.2.a Minimum Formatting and Appearance Requirements

- Title Block – Each sheet shall include a title block with the unique model number of the system, the drawing number, and the date.
- Sheet Numbers – Include sheet numbers for every sheet, e.g. Sheet XX of YY.
- Page Size – Either 8.5 x 11 inch (letter) size or 11 x 17 inch (ledger/tabloid) size is acceptable.
- Scale – Every sheet shall include a drawing scale (e.g. 1/2" = 1′0″ or 1:10). Not-to-scale (NTS) drawings are not acceptable. Note: Section views with scales different than the base view are allowable on sheets with the base view as long as everything is clearly marked.
- Revision Block – Every sheet that includes a revision must have a revision block that states the revision number, revision date, draftsman's initials, and a brief description of what changed.
- Documentation shall be professional in appearance. Rough sketches are not acceptable.

1.8.2.b Minimum Bill of Material (BOM) Requirements

- The BOM should include all of the system's parts.
- The item information provided shall include: item numbers, quantities, short item descriptions (two to four words, e.g. “hinge jamb”), material grades (e.g. ASTM A514), and any applicable manufacturer designations.
- Applicable material specifications (e.g. hot-rolled steel – ASTM A35) shall be included for every material in the system.

1.8.2.c Minimum Labelling Requirements

- On all sections, indicate “attack” and “protected” sides as applicable. The front elevation drawing may be indicated as the “attack side” only.
- The manufacturer shall clearly indicate all proprietary information.

1.8.2.d Minimum Dimensioning Requirements

- Each assembly drawing shall show the arrangement and locations of all bolts, fasteners, sealants, gaskets, weep holes, glazing stops, glazing layups (including spacers), and dimensions (including frame size and vision opening).
- Always include dimensions of pieces, the location and type of welds, the size and type of bolts, and glazing layup(s), including material, size(s), spacing, and arrangement.
- All dimensions shall be dual dimensions (metric and imperial units). The metric units shall be printed first, with imperial units in parentheses.

- For mulled systems, assembly drawings depicting the mullions between the systems shall be included.
- If applicable, provide the clear width opening dimensions for Americans with Disabilities Act (ADA) compliance.

1.8.2.e Minimum View Requirements

- On all elevations/views, indicate from which side the elevation/view is taken.
- Provide three-view (front, top, and side) details for each component.
- Provide system-level exploded views.
- Show all parts and all assemblies.

1.8.2.f Minimum Notes Requirements

Include a general notes sheet to specify:

- Welding codes and procedures (Welding Procedure Specifications (WPS)), if applicable
- Paint specifications, if applicable
- Complete specifications for bolts, nuts, washers, and screws
- Tolerances
- Glazing specifications, if applicable

1.8.3 Installation Instructions

The manufacturer's recommended field installation and user instructions shall be included in the submission for certification. The instructions shall include care and maintenance instructions. The instructions shall include repair instructions, such as for door hinge replacement. The installation instructions must include step-by-step photos for each phase of the installation with a detailed narrative for each instruction phase. The instructions shall comply with OBO requirements.

1.8.4 OBO Specification Guarantee

The manufacturer will submit a letter certifying that their system meets or complies with all currently applicable U.S. Department of State Bureau of Overseas Buildings Operations (OBO) specifications. For certified systems, see **Section 1.4.1(f): Failure to Comply with OBO Specifications**.

1.8.5 Test-to-Failure Agreement

The manufacturer shall submit a letter permitting the system to be tested to failure after the certification testing required by this standard has been successfully completed. If DOS exercises the test-to-failure option, DOS will be responsible for the additional testing and reporting costs.

1.9 SCHEDULING

1.9.1 Testing Arrangements

The manufacturer shall be responsible for arranging and scheduling tests which are required by this standard. These tests must be made at an independent facility. All FE/BR testing shall be at a DS/PSP/PSD-approved test facility.

1.9.2 Scheduling Procedure

All scheduling shall be done in accordance with **Section 1.9.2: Scheduling Procedure**.

1.9.2.a Documents and Scheduling

The documents which are required by **Section 1.8: Certification Documentation** shall be submitted by the manufacturer to the Test Director and DS/PSP/PSD at least **30 working days** prior to any scheduled testing.

Because the DOS certification program is very dependent on accurate drawings, testing to this standard shall not begin until DS/PSP/PSD and the test laboratory are both satisfied that the drawings are neat, clear, accurate, complete, well-organized, and in accordance with **Section 1.8.2: Minimum Requirements for Engineering Drawings and Specifications**.

1.9.2.b Test Plan and Scheduling

The Test Director will produce a test plan at least **five (5) working days** prior to testing in accordance with **Section 2.2.1(b) Formal Test Plan** and submit the test plan to DS/PSP/PSD for approval. Upon DS/PSP/PSD approval of the test plan, the Test Director will submit the test plan to the manufacturer in accordance with **Section 1.9.2: Scheduling Procedure** and **Section 2.2.1(c): Scheduling Testing**.

1.9.2.c Test Sample and Scheduling

The test sample shall be submitted to the test facility at least **five (5) working days** prior to any scheduled testing.

Testing to this standard shall not begin until DS/PSP/PSD and the test laboratory are satisfied that the sample complies with the requirements of **Section 1.7.1: System Sample**.

1.9.2.d Witnessing and Scheduling

All certification testing shall be witnessed by a representative of DS/PSP/PSD. The Test Director shall notify DS/PSP/PSD at least **15 working days** in advance of any scheduled FE/BR testing (see also **Section 2.2.1(c): Scheduling Testing** and **Section 2.7.1: DS/PSP/PSD Shall Witness Testing**). DS/PSP/PSD must approve the test schedule in advance of testing.

1.10 REPORTING

Once a system or device has been tested according to this standard for official DOS approval, a report of all testing conducted on the system must be provided, regardless of the outcome, to DS/PSP/PSD at the address in **Section 1.6: Point of Contact for this Standard** within **30 days** of test completion in accordance with **Section 4.2.5: Test House Test Report**.

1.11 TESTING AND CERTIFICATION OF DOS-DESIGNED FE/BR SYSTEMS

All DOS-designed FE/BR systems which require certification to this standard will comply with all requirements herein, amended as follows:

1.11.1 Sample Size

DS/PSP/PSD must approve the size(s) of the test sample(s), and the size(s) must follow **Section 1.7.1: System Sample** size guidance.

1.11.2 Effect on DOS Model Number

“DOS” shall be used instead of the manufacturer’s three-letter symbol for the system’s DOS Model Number.

1.11.3 Mounting Clearance for DOS-Designed FE/BR Systems

Be in accordance with **Section 1.3.8: Mounting Gap**. If a mounting gap is not required, such as for walls, DS/PSP/PSD must approve the elimination of the gap.

1.11.4 Non-Standard DOS-Designed Systems

Concerning **Section 3.3.2: Concentrated Assault Test**, DS/PSP/PSD shall establish the testing criteria for walls and other specialized FE/BR systems. Tools, test sequencing, and testing location shall follow the intent of **Section 3.3.2** but may be modified to ensure FE/BR criteria are met for nonstandard systems.

1.12 MODEL NUMBERING CONVENTION

1.12.1 DOS Model Number Assignment Procedure

All systems certified under this standard shall be assigned a DOS Model Number according to the convention in **Figure 1: DOS Model Numbering Convention**.

The Test Director shall ensure that the unique model number for every tested system is clearly applied to all documentation submitted with the certification testing report(s) (see **Section 4.0: Test Data and Reporting**), using the manufacturer’s model number, as well as a DOS Model Number if one has been assigned, so there is no confusion over what was tested.

If and only if all of the requirements of this standard (testing, documentation, etc.) have been satisfied, then the new DOS Model Number will be assigned by the U.S. State Department at the time that the Certification Letter is issued.

If the system is being certified under **Section 1.2.5(a): New Certification of Previously Certified Systems**, then it shall receive a new DOS Model Number. This is also stated in **Section 1.2.5(a)**.

1.12.2 Modification Numbering

If a certified system is changed in design or material and is certified under this standard, then the DOS Model Number shall be postscripted with the next consecutive letter (e.g. ‘A’ would become ‘B,’ ‘B’ would become ‘C,’ etc.).

1.12.3 Unique Model Numbers Requirement

Model number sharing is not allowed. Each certified system must have a unique DOS Model Number by which to identify it and differentiate it from all other systems. Manufacturer model numbers shall also be unique, not shared.

2.0 TEST PREPARATION

2.1 TEST DIRECTOR

2.1.1 Test Director Assignment

A Test Director will be assigned to each test. The test facility will assign the Test Director.

2.1.2 Test Director Shall Ensure Compliance

The Test director assigned under **Section 2.1.1: Test Director Assignment** shall ensure, at a minimum, the following:

- For Ballistic Testing
  - Ammunition is provided per **Table 1: Ballistic Resistance Test Ammunition**.
  - The ballistic testing requirements of **Table 6: Minimum Ballistic Test Requirements** are observed.
- For Forced Entry Testing
  - Only authorized tools are employed (see **Table 2: Forced Entry Test Resources**).
  - The required iterative times are met (**Tables 3: Forced Entry Testing of Individual Systems**, **4: Forced Entry Testing of Mulled Systems**, and **5: Forced Entry Testing of Double Door Systems**).
  - The Test Team is selected in accordance with **Section 2.5: Forced Entry Test Team**.
- For Any Testing to this Standard
  - Design or other observed weakness(es) of the test sample is (are) consistently exploited.
  - The test sample and its engineering drawings and specifications are analyzed before the test.
  - All test data is collected and recorded.
  - All reasonable safety precautions are employed during testing (see also **Section 3.7: Safety**).
  - Before a sample is tested, all system hardware, to include lock assemblies, function in the manner intended by the manufacturers.

2.2 TEST SAMPLE DOCUMENTATION AND TEST PLAN

Testing will not be initiated until the test sample documentation is complete and correct (see also **Section 1.9.2: Scheduling Procedure**).

2.2.1 Test Director Shall Examine Test Sample Documentation, and Formulate and Submit a Test Plan

2.2.1.a Examination

The Test Director shall examine the test sample documentation and confirm its completeness and accuracy (see **Section 1.8: Certification Documentation**).

2.2.1.b Formal Test Plan

Upon receipt of the proper drawings and documentation specified in **Section 1.8: Certification Documentation**, the Test Director will identify which dissimilar portions of the system require testing. From these determinations, a formal test plan will be produced in accordance with **Section 1.9.2(b): Test Plan and Scheduling**.

The Test Director will submit the formal test plan to DS/PSP/PSD for approval. DS/PSP/PSD will review the formal test plan to determine if any dissimilar portion may forgo testing due to previous testing or certification, and to ensure that all potential vulnerabilities are investigated.

2.2.1.c Scheduling Testing

Upon approval of the test plan by DS/PSP/PSD, the Test Director will submit the formal test plan to the manufacturer, after which test scheduling will be finalized in accordance with **Section 1.9.2: Scheduling Procedure**. DS/PSP/PSD is given notice in accordance with **Section 1.9.2(d): Witnessing and Scheduling** so that DS/PSP/PSD may arrange for representation (see also **Section 2.7.1: DS/PSP/PSD Shall Witness Testing**).

2.2.2 Test Sample Documentation Examination by Test Team

Forced entry Test Team may not examine test sample documentation once testing has commenced. The manufacturer may request that the Test Team not have access to the sample documentation at all.

2.3 TEST SAMPLE MOUNTING

Test samples shall be rigidly mounted in accordance with **Section 2.3: Test Sample Mounting** and **Section 1.3.8: Mounting Gap**. Test samples shall be tested in accordance with **Section 1.3.8: Mounting Gap** and **Section 3.0: Test Procedures**.

2.3.1 Ballistic Resistance Test Sample Mounting

Samples for ballistic testing only must be rigidly mounted in a manner appropriate to the configuration and purpose of the sample. For samples that will undergo concurrent FE/BR testing (see **Section 1.3.3: Concurrent Testing**), follow **Section 2.3.2: Forced Entry Test Sample Mounting**.

2.3.2 Forced Entry Test Sample Mounting

2.3.2.a Forced Entry Test Fixture

The test sample shall be mounted in a test fixture which is constructed so that its resistance to deformation and distortion is greater than the resistance of the system.

It is recommended that the test fixture be constructed of 1/2″ (13 mm) thick structural steel components.

The test fixture shall simulate installation in a permanent steel or concrete structure which neither enhances nor degrades the forced entry and/or ballistic resistance of the system.

2.3.2.b Forced Entry Test Sample Installation
The test sample shall be mounted in accordance with the manufacturer's instructions, with attention paid to the attack and protected side orientation.

The test frame mounting must give no leverage advantages or disadvantages over the expected mounting conditions in the field.

If the test system cannot be mounted according to mounting instructions required in **Section 1.8.3: Installation Instructions**, then the test shall not be conducted.

The manufacturer may request that active members of the forced entry Test Team not be involved in the construction and/or installation of the test sample.

2.3.2.c Forced Entry Mounting Clearance
Test samples which require forced entry or concurrent testing shall provide a mounting gap over the entire periphery of the assembly.

2.3.2.d Forced Entry Test Samples Which Require Footers
Test samples shall be erected (including cast in place) on footers and either back braced or capped with a simulated roof or ceiling panel. This is intended to be representative of the actual field condition. The bracing and capping shall not artificially enhance or detract from the FE resistance of the system.

2.3.3 Test Sample Labelling
Attack and protected sides shall be clearly labelled by the test sample's provider. Each test sample shall be labelled with its model number. With regard to FE testing, the test sample shall not be marked to indicate the dimensions necessary for the passage of the pass/fail shape(s) (see **Section 3.4.2: Forced Entry Rejection Criteria**).

2.4 BALLISTIC TEST SET-UP
2.4.1 Rifle
2.4.1.a Test Sample Placement
The plane of the threat surface of the test sample shall be positioned at a minimum of 20 feet (6.096 m) from the muzzle of the gun or test barrel to produce zero-degree (0°) obliquity impacts.

2.4.1.b Velocity Measurement Location
If using velocity screens, then velocity screens shall be positioned at 5 feet (1.525 m) and 15 feet (4.57 m) from the muzzle which, in conjunction with a calibrated elapsed time counter, will be used to determine bullet velocities at 10 feet (3.05 m). See **Section 2.6.1(a): Ballistic Testing Equipment** for alternatives to screens. If using a method other than screens, a measurement shall be recorded at 10 feet (3.05 m) from the test sample.

2.4.1.c Witness Panel Placement

Penetrations will be determined by visual examination of a witness panel (see **Table 6: Minimum Ballistic Test Requirements**) placed 6 inches (150 mm) behind and parallel to the system in accordance with **Section 3.2.3: Inspection of Witness Panel**. The witness panel can be placed up to 5 inches (125 mm) behind and parallel to the system. The witness panel may not be placed further than 6-1/4 inches (159 mm) behind and parallel to the system. Witness panels shall be sized to cover the location of the impact area(s) plus at least 12 inches (305 mm) from the area(s) in every direction in the plane of the witness panel.

2.4.1.d Witness Panel Contouring

With the exception of the witness panel mounting for convoluted passages, witness panels shall be contoured to match the test sample.

2.4.2 Shotgun

2.4.2.a Barrel Placement

Barrel placement must be 8 inches (plus or minus 2 inches) (200 ± 50 mm) from the point of impact. Impact obliquity shall be determined by the Test Director consistent with both the maximum exploitation of the test sample and safety.

2.4.2.b Deal Trays and the Witness Panel

For deal trays, the witness panel will be composed of **Table 6: Minimum Ballistic Test Requirements: Witness Panel d** sheet mounted to the test sample deal tray in the following manner: (1) tape the sheet to the inner edge (concave portion) of the tray, and (2) tape the sheet to the protected side of the transparency at a 45-degree angle (See **Figure 2: Placement of Ballistic Witness Panel for Testing of Deal Trays**).

2.4.2.c Witness Panel for Oblique or Convoluted Passages

For oblique or convoluted passages, the witness panel will consist of **Table 6: Minimum Ballistic Test Requirements: Witness Panel d** sheet mounted on the test fixture so that the witness panel should maintain a 6 inch (150 mm) parallel standoff contoured to conform to the shape (on the protected side) of the test sample.

2.4.3 Option to Waive Velocity Measurements

The DS/PSP/PSD representative may, at his or her sole discretion, waive a velocity measurement (see also **Section 3.2.5: Measuring Shot Velocity**).

Velocity measurements may be waived for oblique shots, shotgun shots, and shots on unusually shaped test samples, as a practical consideration, and/or to avoid undue burden to the test facility.

For all other shots, waiving velocity measurements shall be a last resort if all other practical measurements to capture a valid velocity measurement have been exhausted. The inability to capture reliable velocity measurements is a sufficient reason to justify delaying and/or rescheduling testing.

2.5 FORCED ENTRY TEST TEAM

The Test Director shall assign people to the Test Team. Each of the Test Team members shall meet the requirements of **Section 2.5.1: Test Team Minimum Requirements**.

Under **Section 2.3.2(b): Forced Entry Test Sample Installation**, the manufacturer may request that no members of the Test Team have served in the construction and/or installation of the test sample. If the manufacturer makes this request, then the Test Director shall ensure that the request is honored.

2.5.1 Test Team Minimum Requirements

The Test Team members shall each be:

- (1) Physically fit
- (2) In good health
- (3) Between the ages of 18 and 45 years old
- (4) Not less than 150 lbs. (68 kg) in body weight
- (5) Able to lift and carry a minimum load of 60 lbs. (27.2 kg) alone
- (6) In possession of at least a basic understanding of hand tool usage

2.5.2 Number of Test Team Members

During the test, the number of active members of the Test Team (not including data recorders and supervisors) shall be that specified in **Table 2: Forced Entry Test Resources**.

2.6 AMMUNITION, TOOLS, AND SUPPORT INSTRUMENTATION

2.6.1 Ballistic Testing

2.6.1.a Ballistic Testing Equipment

The following equipment is required:

- Firearms or Test Barrels
    - Firearms or test barrels chambered for the ammunition in **Table 1: Ballistic Resistance Test Ammunition** (see also **Table 6: Minimum Ballistic Requirements**)
- A Velocity Measuring Device – Options include:
    - High-speed video with fiducials
    - Radar
    - Velocity screens
    - Other industry-accepted means for accurate, reliable velocity tracking at DS/PSP/PSD discretion
- Witness Panels

- See **Table: 6: Minimum Ballistic Requirements, Witness Panel sub-table**

2.6.1.b Ammunition Requirements

Ammunition for use in ballistic resistance testing to this standard is listed in **Table 1: Ballistic Resistance Test Ammunition** (see also **Table 6: Minimum Ballistic Requirements**).

2.6.1.b.1 Ammunition Configuration and Performance Standards

The test laboratory will certify:

- Commercial Ammunition
  - Commercial ammunition used in conducting these tests will conform to the configuration and performance standards established for that cartridge by the **Sporting Arms and Ammunition Manufacturers' Institute (SAAMI)**.
- Military Ammunition
  - Military ammunition used in conducting these tests will conform to the configuration and performance standards established by **United States Military Specifications**.

2.6.2 Forced Entry Tools

The tools to be used in forced entry testing to this standard are listed in **Table 2: Forced Entry Test Resources**.

2.6.3 General Support Instrumentation

The following general support instrumentation shall be available to document all testing:

- Timekeeping
  - Chronometer able to track and record time down to the nearest second
- Still Images
  - Camera capable of taking images as described in **Section 4.2.8: Photographic Record of Testing**
- Video Recording
  - Digital format

2.7 DS/PSP/PSD REPRESENTATION

2.7.1 DS/PSP/PSD Shall Witness Testing

DS/PSP/PSD will have its representative(s) present for and witness to all FE/BR testing. Therefore, DS/PSP/PSD shall be given advance notice of scheduled testing as described in **Section 1.9.2(d): Witnessing and Scheduling** and **Section 2.2.1(c): Scheduling Testing**.

2.7.2 Controversies or Questions

Controversies or questions which arise during testing will be resolved by the DS/PSP/PSD representative whose decisions will be final.

3.0 TEST PROCEDURES

3.1 GENERAL TEST PROCEDURES

The requirements and procedures presented herein are the minimum required for certification by the U.S. Department of State.

3.1.1 Geometric Commonality and Physical Construction

The U.S. Department of State reserves the right to forego portions of the test requirements (FE and/or BR) for reasons of geometric commonality within the test sample, such as the identical design of the right and left edges of a wall panel. Any changes to the application of the test criteria must be approved by DS/PSP/PSD representative and documented in the test report (see **Section 4.2.4: Waivers**).

3.1.2 Test Variations

The requirements, procedures, or authorized tools of this standard may be varied to reflect unique configurations of the system to be tested. Any such variation must be approved in writing by DS/PSP/PSD whose guideline for granting approval shall be for conditions or design variations unforeseen in establishing these requirements.

The DS/PSP/PSD representative may add testing to the test plan to ensure that all potential vulnerabilities are investigated (see also **Section 2.2.1(b): Formal Test Plan**).

3.1.3 Data Recording

The data shall be recorded in an appropriate data table and photos shall be taken per **Section 4.2.7: Photographic Record of Testing**.

3.1.4 Concurrent FE/BR Testing

When a test sample is to be tested for both forced entry and ballistic resistance, the ballistic resistance testing shall be conducted first, unless **Section 3.4.2(b): Inherent Shape Advantage for Forced Entry** is applicable. See also **Section 1.3.4: Multiple Samples**.

3.1.5 Door Operability

3.1.5.a Measuring the Pull/Push Force

For measuring the pull/push force, use OBO guidance.

3.1.5.b Recording the Pull/Push Force

For recording the pull/push force, use OBO guidance.

3.1.5.c Option for Door Closer Removal

The door closer and its removal is at the direction of OBO guidance.

3.2 BALLISTIC TESTING

3.2.1 Firing Procedure

Ammunition of the appropriate caliber and type (See **Table 1: Ballistic Resistance Test Ammunition**) shall be single-fired in accordance with the minimum requirements in **Table**

6: Minimum Ballistic Test Requirements. The firings will be conducted to examine the ballistic resistance of:

- (1) All features of the test sample (frame, seams, locks, hinges, etc.).
    - Notes:
        - (1) The manufacturer is responsible for the ballistic resistant protection of the mounting gap between the test sample and the test fixture (see also **Section 1.3.8: Mounting Gap**).
- (2) All variations in the ballistic cross section of the test sample(s).
- (3) Convoluted design passages (deal trays, louvers, passive speak through features, etc.) to angled and ricocheted fire from a 12-gauge shotgun.
    - Note: The shotgun test is a additional test that may be required for convoluted passages at DS/PSP/PSD discretion.

3.2.2 Additional Firings

The number and location of ballistic impacts of **Table 6: Minimum Ballistic Test Requirements** are minimum requirements. The Test Director and/or DS/PSP/PSD representative may require unlimited additional firings, subject to the conditions in **Section 3.2.2(a): Limits on Transparencies**, **3.2.2(b): Limits on Opaque Portions**, and **3.2.2(c): Limits on Ammunition**.

3.2.2.a Limits on Transparencies

The separation between any two ballistic impacts on a single contiguous transparency area shall be no less than 4 inches (100 mm) on center.

3.2.2.b Limits on Opaque Portions

The separation between any two ballistic impacts on a single contiguous area of opaque material or fitting will be no less than 1.5 inches (38 mm) on center regardless of the size of the disturbed area from each impact.

3.2.2.c Limits on Ammunition

Ammunition must meet the performance requirements of **Table 1: Ballistic Resistance Test Ammunition**.

These shots must be made within the constraints of range and the safety of all persons (see **Sections 2.4.1: Rifle**, **2.4.2: Shotgun**, and **3.7: Safety**).

3.2.2.d No Limits on Oblique Angle Shots

There is no limit to the number of oblique-angle shots required to examine all possible penetration paths of convoluted passages. The results of oblique shots will be photographed and diagrammed as necessary to ensure consistent repeatability.

3.2.3 Inspection of Witness Panel

3.2.3.a Classified as "Complete Penetration"

After each firing, the witness panel shall be visually inspected from the protected side by holding a light as bright or brighter than 800 lumens close to the other side to see if light passes through the witness panel. Only a perforation of the witness panel, whether by bullet fragments or material from the test sample (spall), shall be classified as a "Complete Penetration."

3.2.3.b Classified as "No Penetration" or "Partial Penetration"

Impacts which produce no deformation on the strike-face of the sample shall be classified as "No Penetration." Any other results which are not deemed a "Complete Penetration" (**Section 3.2.3(a): Classified as "Complete Penetration"**) or "No Penetration" will be classified as "Partial Penetration". The definition of "Partial Penetration" applies whether or not the test sample has been completely perforated. Ballistic rejection criteria are covered in **Section 3.4.1: Ballistic Rejection Criteria.**

3.2.4 Fair Hits

For this Standard, a "Fair Hit" is any of the following:

- (1) A zero-degree oblique ballistic impact of the specified weight and bullet type within the specified velocity range on the specified location of the sample (see **Table 1: Ballistic Resistance Test Ammunition**) is considered fair.
- (2) An impact at less than the minimum acceptable velocity or at excessive obliquity or with excessive yaw which results in complete penetration and is otherwise a fair hit is considered fair.
- (3) An impact which does not produce complete penetration, but which exceeds the maximum acceptable velocity and qualifies as a fair hit in all other respects is considered fair.
- (4) Any oblique shots to exploit convoluted passages and vulnerable pathways are considered fair.

3.2.5 Measuring Shot Velocity

The velocities of all shots fired in ballistic testing will be measured individually using equipment in **Section 2.6.1(a): Ballistic Testing Equipment** to ensure they meet the velocities specified by **Table 1: Ballistic Resistance Test Ammunition**. Any velocity measurements waived under **Section 2.4.3: Option to Waive Velocity Measurements** must be documented in the test report in accordance with **Section 4.2.4: Waivers**.

3.3 FORCED ENTRY TESTING

3.3.1 General

Forced entry testing shall be conducted on assemblies including, but not necessarily limited to: doors, windows, and wall panels of any type; louvers; escape hatches; and protective window grilles. All forced entry testing, regardless of the type of assembly or level of protection, shall consist of a concentrated assault test of edges and other critical locations (see, as applicable, **Table 3: Forced Entry Testing of Individual Systems**, **Table 4: Forced Entry Testing of Mulled Systems**, and **Table 5: Forced Entry Testing of Double Door Systems**).

3.3.1.a Interruptions

The timed forced entry test shall be conducted without interruption except for: safety (see **Section 3.3.1(b): Forced Entry Testing Safety**), scheduled rests (see **Section 3.3.1(c): Rests**), conference with Department of State observers, tool replacement (see **Section 3.3.1(f): Test Resources**), or technical difficulties (e.g. data collection malfunction involving photo or video capture).

3.3.1.b Forced Entry Testing Safety

The test may be interrupted for reasons of safety (imminent danger to or injury of test attendees including but not limited to Test Team). This time will not be used for clearing away debris, such as glass fragments which have been produced as a result of the testing, from the test sample. Any modifications to the test sample made for safety reasons must be agreed to by all parties and must not in any way enhance or detract from the sample's resistance to forced entry. See also **Section 3.7: Safety**.

3.3.1.c Rests

Rests shall be given only at the end of a test. However, during 60-minute tests, rests shall be given every 15 minutes. These scheduled rests shall have a duration of at least 15 minutes; there is no maximum duration. The Test Team will not use this time: for planning purposes, to discuss attack techniques with each other, to inspect the test sample, or to confer with the Test Director about the progress of the test.

3.3.1.d Inspections

Once testing has commenced, time used by the Test Team to examine the test sample shall be considered part of the test time duration. If it is safe to do, Test Team members may look or reach through the test sample. The Test Director will determine whether the Test Team is able to look or reach through the test sample safely.

3.3.1.e Test Resources

Except when necessary under **Section 3.3.6: Test Sample Repairs and Replacements**, only those tools specified in **Table 2: Forced Entry Test Resources** may be applied to the test sample once forced entry testing has commenced.

The tools shall be well-made, and be in good condition – undamaged, with like-new effectiveness – when forced entry testing commences.

Test resources may not be applied to a test sample during interruptions (see **Sections 3.3.1(a): Interruptions**, **3.3.1(b): Forced Entry Testing Safety**, and **3.3.1(c) Rests**).

Sharp-edged tools shall not be re-sharpened during the test.

Tools which become unserviceable or unsafe during testing may be replaced in accordance with **Table 2: Forced Entry Test Resources – Number of Replacements Available**. Time taken to make this replacement will count as an interruption under **Section 3.3.1(a): Interruptions** and not considered as part of the test time duration.

During testing, if a tool is dropped through an opening in a test sample, and the Test Team can safely reach through the sample to retrieve it, then the Test Team may do so. The Test Director will determine whether the Test Team is able to retrieve the tool safely.

3.3.2 Concentrated Assault Test

The concentrated assault will begin with the area or dissimilar portion predetermined by the Test Director to be most vulnerable to forced entry.

3.3.2.a Test Sequence

This test sequence consists of attempting to force an entry through any portion of the sample – to include, but not limited to, the frame, opaque panels, transparency, transparency mounting, hinges, locks, fittings, mounting devices, etc. – using the tools and Test Team members shown in **Table 2: Forced Entry Test Resources**.

3.3.2.b Test Intent

The intent of this test is to attempt to penetrate the panel and/or transparency portions of the assembly or to distort an edge in order to disengage a fitting or device from its mountings in the test sample. This test will be performed on each dissimilar portion of the test sample.

3.3.2.c Exploit Dissimilar Portions

The Test Director shall select and exploit any dissimilar portion of the test sample using the approved formal test plan (see **Section 2.2.1(b): Formal Test Plan**).

After the first test, an analysis of the test sample may be performed by the DS/PSP/PSD representative and the Test Director to determine if additional tests are necessary on any of the other dissimilar portions of the test sample. Additional tests may be added to the test plan, or tests may be removed from the test plan. Any tests not performed will be documented in the test report (see also **Section 4.2.4: Waivers**).

3.3.2.d Timing

Given that the forced entry test is timed, the Test Director shall impress the Test Team with a sense of urgency.

The test times shall be in accordance with **Table 3: Forced Entry Testing of Individual Systems**, **Table 4: Forced Entry Testing of Mulled Systems**, and/or **Table 5: Forced Entry Testing of Double Door Systems**, as applicable. See **Section 3.3.3: Mulled**

Systems for test time guidance for mulled systems. See **Section 3.3.4: Double Door Systems** for test time guidance for double doors.

Periods of non-activity (see **Sections 3.3.1(a): Interruptions**, **3.3.1(b): Forced Entry Testing Safety**, and **3.3.1(c): Rests**) are not to be charged to the required test times.

3.3.2.e Progress Information

During periods of non-activity (see **Sections 3.3.1(a): Interruptions**, **3.3.1(b): Forced Entry Testing Safety**, and **3.3.1(c): Rests**):

- (1) The Test Director will ensure that the Test Team does not discuss the progress of the concentrated assault.
- (2) The Test Director will ensure that Test Team does not inspect the test sample.
- (3) The Test Director will not discuss the progress of the concentrated assault with Test Team.

3.3.2.f Stepping Behind the Test Sample

No member of the Test Team may step behind the test fixture, or behind a plane parallel and contiguous to the face of the attack side of the test sample once forced entry testing has commenced. See also **Section 3.3.1(d): Inspections** regarding a tester's ability to look or reach through a test sample, and **Section 3.3.1(e): Test Resources** regarding a tester's ability to retrieve a lost tool.

3.3.3 Mulled Systems

Forced entry testing of test samples consisting of two or more systems mulled to one another shall be conducted in accordance with **Table 3: Forced Entry Testing of Individual Systems** if the systems are mulled directly together, or in accordance with **Table 4: Forced Entry Testing of Mulled Systems** if the systems are mulled with separate mullions between the systems.

Note: Seismic joints are to be considered as mulled connections.

3.3.4 Double Door Systems

Forced entry testing of double door systems, to include the door framing, shall be conducted in accordance with **Table 5: Forced Entry Testing of Double Door Systems**.

3.3.5 Mounting Gap Forced Entry Testing

The mounting gap shall be tested per **Section 3.3: Forced Entry Testing** for the specified protection levels (see, as appropriate, **Table 3: Forced Entry Testing of Individual Systems**, **Table 4: Forced Entry Testing of Mulled Systems**, and/or **Table 5: Forced Entry Testing of Double Door Systems**). Also, see **Section 1.3.8: Mounting Gap**.

3.3.6 Test Sample Repairs or Replacements

3.3.6.a During or Between Testing

If a sample is being used for both ballistic and forced entry testing, then no repairs or replacement of damaged components are permissible during or between the ballistic resistance testing and any of the forced entry testing.

No repairs or replacement of damaged components are permissible during or between any of the forced entry tests on a sample.

3.3.6.b After Testing

After the completion of the Concentrated Assault Test with respect to one dissimilar edge, the Test Director, with the approval of the DOS representative, may direct limited repairs to features which have been completely evaluated if he judges the repairs necessary to fairly evaluate the yet untested area. Any such repair must not enhance or detract from the forced entry resistance of the untested area. All of these repairs shall be documented in the test report.

3.4 PASS/FAIL CRITERIA

The results of testing in accordance with this standard will be evaluated as described in **Section 3.4: Pass/Fail Criteria**.

3.4.1 Ballistic Rejection Criteria

Any penetration deemed a “complete penetration” under **Section 3.2.3(a): Classified as “Complete Penetration”** by a hit deemed a “fair hit” under **Section 3.2.4: Fair Hits** is cause to reject the design of that sample for the level of protection tested, regardless of the quantity of other shots deemed “no penetration” or “partial penetration” under **Section 3.2.3(b): Classified as “No Penetration” or “Partial Penetration**.**”**

3.4.2 Forced Entry Rejection Criteria

3.4.2.a Pass/Fail Shapes

Any failure of the manufacturer’s recommended mounting hardware or penetration of any portion of the system sufficient to permit full passage of either pass/fail shape described in this section using manual (hands only) light pushing force within the times of the forced entry Concentrated Assault Test shall be cause to reject that sample design for the level of protection tested:

- Rectangle – A rigid rectangular shape, the exterior dimensions of which measure 12″ x 12″ x 8″ (305 x 305 x 200 mm)
- Cylinder – A rigid cylinder, the exterior diameter of which is 12″ (305 mm), and the height of which is 12″ (305 mm)

Both pass/fail shapes above must be present for the duration of Forced Entry testing.

3.4.2.b Inherent Shape Advantage for Forced Entry

Systems of a shape that is inherently not vulnerable to the pass/fail shapes will be evaluated on a case by case basis by the Department of State.

If the system's dimensions are inherently not vulnerable to the pass/fail shapes in **Section 3.4.2(a): Pass/Fail Shapes** then the system will not automatically earn a passing FE rating.

Further, if there is an inherent shape advantage, then the forced entry portion of certification testing may be required before the ballistic portion of certification testing under **Section 3.1.4: Concurrent FE/BR Testing.** For example, a gun port with an opening smaller than the pass/fail shapes may be required to undergo FE testing before it undergoes ballistic testing, and any opening created during that FE testing which could permit bullets to go through could be sufficient reason to fail the system's ballistic performance.

3.4.3 Certification Criteria

Any system not rejected under **Section 3.4.1: Ballistic Rejection Criteria** or **Section 3.4.2: Forced Entry Rejection Criteria** of this standard, and which also satisfies all of the requirements of **Sections 1.7: Test Sample** and **Section 1.8: Documentation**, will be eligible for DOS certification.

3.5 TEST TO FAILURE

After any system undergoes testing in pursuit of FE and/or BR certification under this standard, the Department of State may require additional testing with the same or additional tools to determine the limits of protections afforded by a system. It may also test the limits of a feature or subsystem of any system which failed to satisfy the requirements for certification. No such testing will be initiated until certification testing of the test sample has concluded.

3.6 DISPOSITION OF TEST ASSEMBLY

The manufacturer is responsible for the final disposition of the test sample. After completion of all testing (including Test-To-Failure under **Section 3.5: Test to Failure**, if appropriate) the manufacturer may abandon the systems to the testing facility or direct the return of the systems to the custody of the manufacturer.

3.7 SAFETY

3.7.1 Safety Considerations

Consider things like: eye protection, hearing protection, inhalation of test-generated debris, slip and fall, tool misuse, broken tools, biohazard containment (i.e. in case of an injury to a Test Team member), and dropped tools. The preceding list is not a complete or exhaustive list of all considerations necessary for testing to this standard. See also **Section 3.3.1(b): Forced Entry Testing Safety**.

3.7.2 Responsibility for Safety

The standard does not purport to address all of the safety concerns associated with its use.

It is the responsibility of the testing facility to establish appropriate safety and health practices, including Personal Protective Equipment (PPE) policies, and to determine the applicability of regulatory limitations. Proper precautions should be taken by the test facility to protect workers, observers/witnesses, and the community.

It is the responsibility of the Test Director to ensure the safety of all persons involved in testing to this standard. It is the responsibility of the Test Director to adhere to all applicable local and federal regulations. It is the responsibility of the Test Director to enforce the testing facility’s practices and the applicable regulatory limitations. The Test Director is responsible for safety and will ensure that all reasonable safety precautions are employed.

4.0 TEST DATA AND REPORTING

4.1 GENERAL REPORTING

Once a system has been tested according to the certification requirements of this standard, a final FE/BR test report of the results of all testing shall be submitted to DS/PSP/PSD by the DS/PSP/PSD-approved test facility for each tested system, regardless of the testing outcome(s). All test reports as well as any information concerning the results of each test are considered the proprietary information of the manufacturer.

4.2 MINIMUM REPORTING REQUIREMENTS

The reporting requirements presented here are the minimum to support certification to this standard. At DS/PSP/PSD discretion, additional reporting may be required for certification to this standard.

4.2.1 All-Inclusive

The final FE/BR test report shall be an all-inclusive document that includes: the configuration of the system, the data and results of all testing, and all of the other documentation required by **Section 4.2: Minimum Reporting Requirements**.

4.2.2 Report Title

The title of the final FE/BR test report shall indicate the type of report, i.e. "Test Report," and the manufacturer's model number of the system including its category and type, i.e. "15-min FE and Rifle BR" (see **Sections 1.3.1: Ballistic Resistance (BR)** and **1.3.2: Forced Entry (FE) Resistance**).

4.2.3 Configuration Documentation

4.2.3.a Complete Configuration Documentation Required

The final FE/BR test report shall contain complete configuration documentation suitable for easy viewing when printed using a standard office printer (see **Section 1.8.2: Minimum Requirements for Engineering Drawings and Specifications**).

4.2.3.b Verification of Identical System

Verification by the test facility shall be included with the final FE/BR test report that the test sample was identical to the system shown in the manufacturer's submitted drawings. Any comments concerning inconsistencies between the assembly and its documentation shall be provided in the final FE/BR test report.

4.2.4 Waivers

The final FE/BR test report shall include any applicable DOS concession of any requirement of this standard in the report's narrative summary section (see **Section 4.2.9: Narrative Summary**). Examples of potential waivers are found in **Sections 2.4.3: Option to Waive Velocity Measurements**, **3.1.1: Geometric Commonality and Physical Construction**, and **Section 3.3.2(c): Exploit Dissimilar Portions**.

4.2.5 Test House Test Report

The final FE/BR test report from a test facility for any system which has been tested to this standard shall specify: the system tested, the as-tested size of the system, the as-tested orientation of the system relative to the horizontal axis, the as-test size of the mounting gap, the model numbers of other assemblies to which the system has been mulled (if applicable), the testing performed, and all test results.

4.2.6 Ballistic Test Data

The type, velocity, impact location, and result of each shot (see **Section 3.2: Ballistic Testing**) shall be included with the final FE/BR test report.

4.2.7 Forced Entry Test Data

Detailed data records of forced entry testing including tools, times, model number(s) of other mulled assemblies (if any), and results (see **Section 3.3: Forced Entry Testing**) shall be included with the final FE/BR test report.

4.2.8 Photographic Record of Testing

Each photo shall be:

- 1) Digital
- 2) In color
- 3) In focus – Crisp and clear
- 4) In high-resolution, suitable for printing, and at least 8 megapixels
- 5) Of a common file format
- 6) Clearly labelled or accompanied by a description which includes the view, the time, the test, and the manufacturer

At a minimum, include the following photos with the final FE/BR test report:

- Photos of the test sample before testing, of both the attack side and the protected side.
- Photos after ballistic testing, for each round, as well as for any shot classified as a "Complete Penetration" under **Section 3.3.2(a): Classified as "Complete Penetration."**
- Photos after each phase of the forced entry testing, of both the attack side and the protected side. Pictures shall include a reference for scale, i.e. a ruler or tape measure.
- For forced entry tests, photos at each rest break, of both the attack side and the protected side (if applicable). Pictures shall include a refence for scale, i.e. a ruler or tape measure.
- At the point at which entry is forced (if applicable).

4.2.9 Narrative Summary

Included with the FE/BR final test report shall be a narrative summary including: the identity of the test facility and any test witnesses, a description of the sample(s), a description of the testing, a description of the results, and a detailed explanation of the basis for any DOS-approved waivers.

4.2.10 Video Recording of Testing

Video footage of testing to this standard shall:

- 1) Be recorded in a common digital format which DS/PSP/PSD is capable of accepting and reviewing
- 2) Include appropriate narrative description and comments
- 3) Be included as part of the final FE/BR test report documentation

Note: Video recording is not required for the ballistic resistance portion of the testing

Entire forced entry testing sequences shall be recorded, the timer shall be visible in the videos. Record video footage of all forced entry testing from both the attack side and the protected side. A watermark shall be included in the video footage including the: date, time, manufacturer, and all applicable model numbers.

4.3 CERTIFICATION ISSUANCE

If all of the requirements of this standard are met, and OBO verifies that all OBO requirements are met, then a certification letter will be issued by DS/PSP/PSD to the manufacturer for that system (see also **Section 1.2: Certification**).

5.0 TECHNICAL REVIEW

5.1 CERTIFICATION BY TECHNICAL REVIEW (T/R)

Certification may be granted based on a Technical Review of the system, its full disclosure drawings and specifications (see **Section 1.8.2: Engineering Drawings and Specifications**), and all test reports for each similar previously tested and/or previously certified system. New testing and/or retesting may be required under **Section 5.3.3: New Testing and/or Retesting**.

Technical Reviews are granted at the discretion of DS/PSP/PSD on a case by case basis. DOS reserves the right to grant or reject a request for certification.

Certifications by Technical Review are limited to only the situations in **Section 5.1: Certification by Technical Review** described below.

5.1.1 Change to Certified System

- Modifications or improvements are being made to certified systems.
- This includes modifications that are necessary in order to meet OBO specifications.
- A new or different size, orientation, shape, or aspect ratio that falls outside the range of the original certification.

5.1.2 Integration of Subassemblies of Certified Systems

- Two or more subassemblies from certified systems are being integrated.

5.1.3 Upgrade to Previously Tested System

- Previously tested non-certified systems are being modified by material or design upgrade.

5.1.4 Request for New Certification Under Current Version of this Standard

- At the discretion of the Department of State, a previously certified system is being newly certified under the current version of this standard in accordance with **Section 1.2.5(a): New Certification of Previously Certified System**.

5.2 REQUESTS FOR CERTIFICATION BY TECHNICAL REVIEW

A request for certification by a manufacturer which is based on this section of the Standard must include the following, at a minimum:

- 1) Drawings and Specifications
    - Include a complete, current set of full disclosure drawings and specifications for the candidate system in accordance with **Section 1.8.2: Engineering Drawings and Specifications**
    - Include a full set of drawings and specifications for each relevant previously certified and/or tested system in accordance with **Section 1.8.2: Engineering Drawings and Specifications**.

- 2) Narrative Explanation
    - Include an explanation of the merits of the request.
    - Clearly explain each difference between previously certified and/or tested system(s) and the candidate system.
- 3) Ratings
    - Include a statement of which rating(s) are sought under this request for Technical Review.
- 4) Scope
    - Include a statement of whether the request applies to a single unit, to a single production lot, or to all units produced after certification date.
- 5) Other Documents Required by **Section 1.8: Certification Documentation**
    - Include installation instructions in accordance with **Section 1.8.3: Installation Instructions**.
    - Include an OBO Specification Guarantee in accordance with **Section 1.8.4: OBO Specification Guarantee**.
- 6) Unique Model Number
    - Include change(s) in model number(s) to comply with **Section 1.12: Model Numbering Convention**.

5.3 TECHNICAL REVIEW PROCEDURE

This section describes the procedure for when a candidate system is eligible for Technical Review under **Section 5.1: Certification by Technical Review (T/R)** and DS/PSP/PSD grants the request.

5.3.1 Compare Drawings and Specifications

The Technical Review by DS/PSP/PSD will compare the drawings and specifications of the candidate system with the drawings and specifications of the manufacturer's previous system(s).

5.3.2 Physical Inspection

A physical inspection of the candidate system may be required.

5.3.3 New Testing and/or Retesting

New and/or retesting at a DS/PSP/PSD-approved test facility may be required at the discretion of DS/PSP/PSD in order to fully evaluate the Technical Review.

5.3.4 Final Decision

5.3.4.a Approval

If DS/PSP/PSD approves a Technical Review, then DS/PSP/PSD shall issue the appropriate certification letter as described in **Section 4.3: Certification Issuance** to the manufacturer.

5.3.4.b Rejection

If DS/PSP/PSD rejects a Technical Review of the candidate system, then the manufacturer may submit a new request for certification via new or further testing to DS/PSP/PSD.

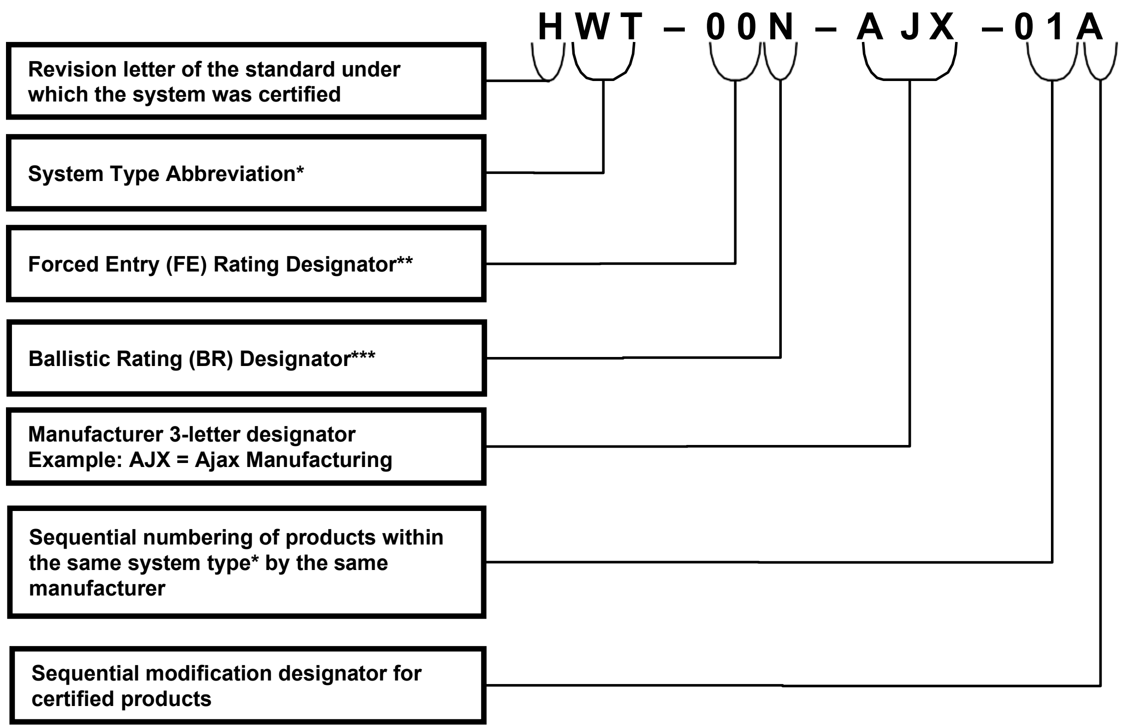
See TAB 2: Figures

See TAB 3: Tables

See TAB 4: Forms

TAB 2: FIGURES

Figure 1: DOS Model Numbering Convention



*See **Table 9:DOS Model Numbering Abbreviations for System Type**

FE Rating Name	FE Designator
None	00
5-Minute	05
15-Minute	15
60-Minute	60

See list above, and see **Section 1.3.1(a): Forced Entry Resistance Levels

BR Rating Name	BR Designator
None	N
Rifle	R
Armor-Piercing	AP
Shotgun	SH

***See list above, and see **Section 1.3.2(a): Ballistic Resistance Levels**

Figure 2: Placement of Ballistic Witness Panel for Testing of Deal Trays

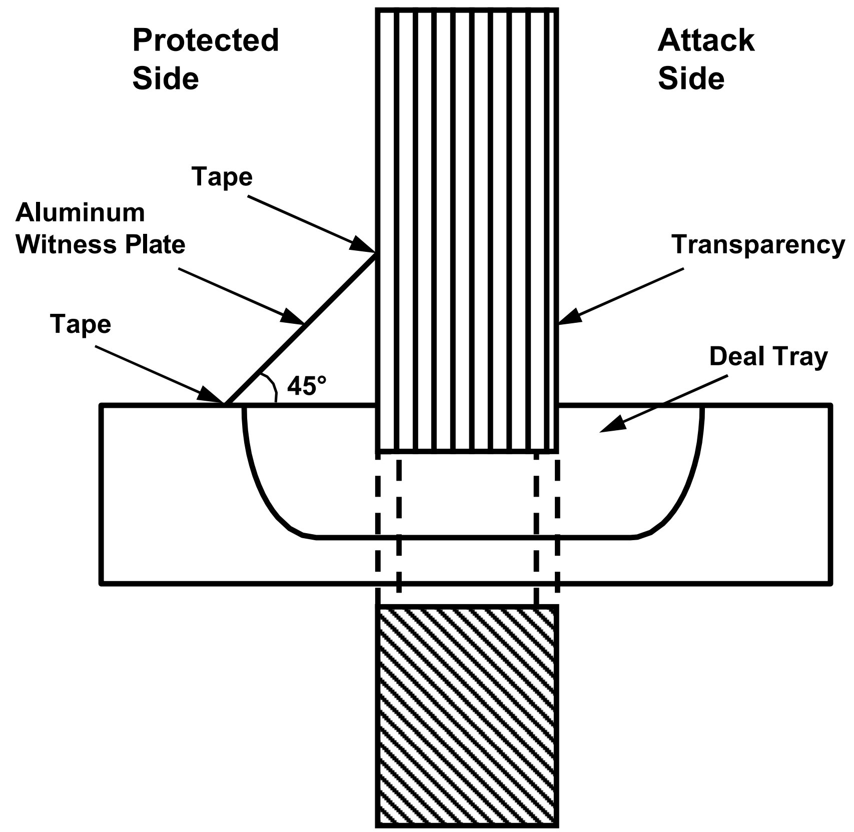
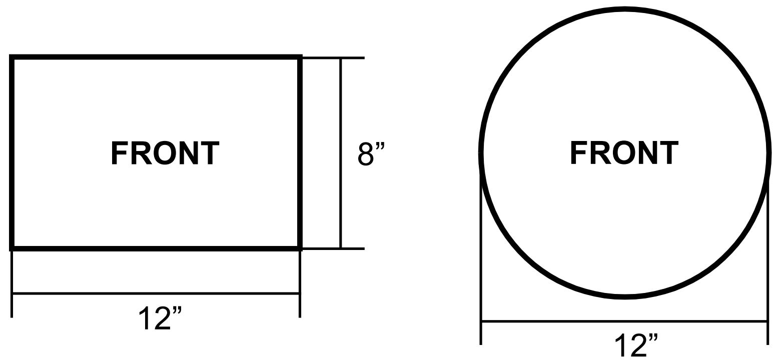


Figure 3: FE Pass / Fail Shape

A rigid rectangular shape with outer dimensions measuring 12” x 12” x 8” (305 x 305 x 200 mm) or a rigid cylinder with outer dimensions of 12” x 12” (305 x 305 mm). The height of the shapes shall be 12” (305 mm). Side marked “FRONT” must go through the sample first. Both shapes shall be provided to the attack team and available for use throughout all FE testing. Neither shape shall weigh more than 20 lbs. (9 kg.)

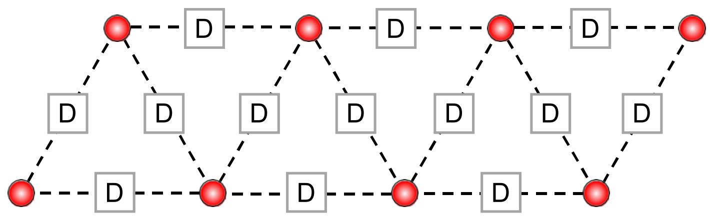


Drawn To Scale
Scale: 1” = 8”

Shape	Perimeter		Width		Length		Area	
	in.	mm	in.	mm	in.	mm	in.^{2}	mm^2
Rectangle	40	1,010	8	200	12	305	96	61,000
Cylinder	37-$^{11}/_{16}$	958	12	305	12	305	113-$^{1}/_{8}$	73,062

Figure 4: 8-Shot Shot Spacing

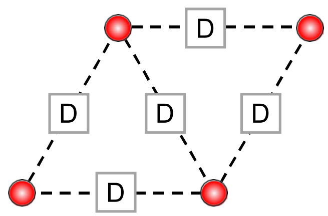
Shoot this pattern for each round. Rounds may be shot in any order but one must complete the pattern before starting with the next round. The shot sequence should be such that debris from prior shots does not enhance the BR performance of the sample. See also **Section 3.2.2(a): Limits on Transparencies** and **Section 3.2.2(b): Limits on Opaque Portions**.



Drawn To Scale
Scale: 1” = 6-$^{7}/_{8}$”
D = 180 mm ± 13 mm
(6-$^{7}/_{8}$” in ± 0.5 in)

Figure 5: 4-Shot Shot Spacing

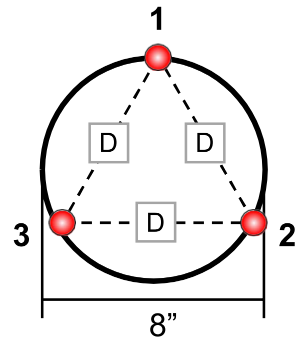
The shot sequence should be such that debris from prior shots does not enhance the BR performance of the sample. See also **Section 3.2.2(a): Limits on Transparencies** and **Section 3.2.2(b): Limits on Opaque Portions**.



Drawn To Scale
Scale: 1” = 6-$^{7}/_{8}$”
D = 180 mm ± 13 mm
(6-$^{7}/_{8}$” in ± 0.5 in)

Figure 6: Historical Rev. G 3-Shot Pattern

Historical Revision G Shot Pattern:
120° ±5° intervals on the periphery of an 8” (205 mm) diameter circle.



Drawn To Scale
Scale: 1” = 6-$^{7}/_{8}$”
D = 6-$^{7}/_{8}$” in ± 0.5 in
(180 mm ± 13 mm)

TAB 3: TABLES

This page intentionally left blank.

Table 1: Ballistic Resistance Test Ammunition

Ballistic Rating	Ammunition Description	Ammunition Identity	Bullet Weight (grain)	Minimum Velocity mps (fps)	Maximum Velocity mps (fps)
Rifle, Military (R) (c)	7.62x51mm, NATO Ball, Steel Jacketed Lead Core, FMJ	M80	147 gr.	823.0 mps (2700 fps)	853.5 mps (2800 fps)
	5.56x45mm, M193, Copper Jacket Lead Core, FMJ	M193	55 gr.	955.5 mps (3135 fps)	986.0 mps (3235 fps)
	5.56x45mm, Copper Jacketed Steel and Lead Core, FMJ	M855	63 gr.	899.2 mps (2950 fps)	929.6 mps (3050 fps)
Rifle, Armor Piercing (AP) (a) (c)	.30-06 Caliber Rifle, Copper Jacketed Steel Core, APM2, (AP)	Original Ammunition Design	166 gr.	838.2 mps (2750 fps)	868.7 mps (2850 fps)
Shotgun (SH) (b) (c)	12 ga. 70mm (2 ¾ inch), #4 Buckshot,27 Pellet, Pellet Diameter 6mm (0.24 inch)	N/A	20.6 gr. per pellet	N/A	N/A

mps = Meters Per Second, fps = Feet Per Second

Table 1 Notes	
(a)	Optional, additional ballistic threat reserved for special products
(b)	Required for all samples with unobstructed, convoluted ballistic paths; the formal test plan (see **Section 2.2.1(b): Formal Test Plan**) shall describe the shotgun testing, but final discretion about the location and number of shots resides with the DS/PSP/PSD representative
(c)	Chronographing may be waived for extremely oblique firings

Table 2: Forced Entry Test Resources

Resource	Description	Quantity of Each for the Specified FE Protection Level in Play			Number of Replacements Available
		5-Minute	15-Minute	60-Minute	
Active Personnel	See **Section 2.5.1: Test Personnel Minimum Requirements**	6	6	6	2
Sledgehammer	10 lbs. (4.54 kg), ≥30 in. long	4	4	4	4
Carpenter Hammer	3 lbs.	2	2	2	1
Carpenter Hammer	1 lb.	2	2	2	1
Two-Man Ram	120 lbs., 2 man, 4 x 4 impact area (±0.25 in.), symmetrical weight distribution	1	1	1	1
Wood-Splitting Maul	9 lbs., 35 in. long	1	1	1	1
Hatchet	≤ 1.5 lbs., ≥14 in. long	1	1	1	0
Medium Axe (e.g. Fiskars X11 Axe, AKA Splitting Axe)	17 in.	1	1	1	0
Pick/Mattock	5 lbs., 36" handle	1	1	1	1
Crowbar, Pinch Bar	60 in.	2	2	2	2
Crowbar, Ripping Bar	48 in.	2	2	2	2
Crowbar, Ripping Bar	24 in.	2	2	2	2
Metal Wedge, Wood Splitting (b)	9 x 2 1/2 in.	4	4	4	2
Standard Hacksaw	12 in., 24 TPI	2	2	2	2
Mini hacksaw	10 in., compact grip, 24 TPI	2	2	2	2
Hacksaw Replacement Blades	24 TPI	6	6	6	0
Wood Hand Saw	12 in.	1	1	1	1
Keyhole Saw	12 in.	1	1	1	1
Bolt Cutters	42 in.	1	1	1	1
End Nippers	14 in.	1	1	1	1
Chisel, Cold (b)	1 in.	2	2	2	2
Chisel, Cold (b)	3/4 in.	2	2	2	2
Chisel, Masonry (b)	2 1/4 in.	2	2	2	2
Screwdriver, Flat Blade (AKA slothead or flat head)	10 in., No. 6	2	2	2	2
Screwdriver, Flat Blade (AKA slothead or flat head)	10 in., No 4	2	2	2	2
Screwdriver, Phillips	10 in., No. 3	2	2	2	2
Screwdriver, Phillips	10 in., No. 1	2	2	2	2
Channel Lock Pliers	10 in.	1	1	1	1
Wrench, Adjustable	15 in.	1	1	1	1
Wrench, Adjustable	10 in.	2	2	2	1
Pipewrench	24 in.	1	1	1	1
Punch, Flat	3/8 in.	1	1	1	1
Punch, Flat	1/4 in.	1	1	1	1

Resource	Description	Quantity of Each for the Specified FE Protection Level in Play			Number of Replacements Available
		5-Minute	15-Minute	60-Minute	
Jaw Locking Pliers (Vice Grip)	12 in.	1	1	1	1
Push Broom	Wooden, 24-28 in.	1	1	1	1
Lumber	2 x 4 x 72 in	2	2	2	0
Wood Cribbing (AKA wood blocks)	6 x 6 x 16 in. kiln-dried Pine	2	2	2	0
Wood Cribbing (AKA wood blocks)	4 x 4 x 16 in. kiln-dried Pine	4	4	4	0
Steel Pipe, Round	2 x 60 in., Schedule 40	2	2	2	1
Steel Tube, Square	1/8 in. thick x 2 in. x 2 in. x 72 in., A500	2	2	2	1
SAE Socket Set with Ratchet	1/4 to 1 1/8 in. min. range	1	1	1	1
Metric Socket Set with Ratchet	6 to 30 mm min. range	1	1	1	1
SAE Hex Key Set	5/64 to 3/8 in. min. range, long arm (e.g. 2-7/8 in to 7-1/8 in.), ball end is not acceptable	1	1	1	1
Metric Hex Key Set	1.5 to 8 mm min. range, long arm (e.g. 73 to 181 mm), ball end is not acceptable	1	1	1	1
Retractable Utility Knife (AKA box-cutter)	Metal handle, metal casing	1	1	1	1
Replacement Blades for Utility Knife	Standard-size removable utility knife blades, heavy duty, made of 0.024 in. premium grade carbon steel	25	25	25	0
Pocket Knife	3.25 in. (83 mm) min. length blade, 5 in. (127 mm) max. length blade, .12 in. (3 mm) blade thickness	1	1	1	1
Rope	Nylon 3-strand, 1/2 in. dia., 20 ft. length	1	1	1	0
Metal Hand File	8 in.	1	1	1	1

Table 2 Notes	
*	These tools are sizes commonly available in the U.S. For SI (metric) equivalents, convert as follows: one (1) in. = 25.4 mm; one (1) lb. = 0.454 kg. Round in accordance with **FED-STD-376A, *Preferred Metric Units for General Use by the Federal Government, 5 May 1983***.

Table 3: Forced Entry Testing of Individual Systems

Concentrated Assault Test	Specified Protection Level		
	5 Minutes	15 Minutes	60 Minutes
Required Testing Time (Minutes):			
Most Vulnerable Location (a)	5	15	60
Additional Selected Locations (a)	5	15	60

Table 3 Notes	
(a)	See **Section 3.3.2: Concentrated Assault Test**

Table 4: Forced Entry Testing of Mulled Systems

Concentrated Assault Test	Specified Protection Level		
	5 Minutes	15 Minutes	60 Minutes
Required Testing Time (Minutes): Each mullion (a) (b)	5	15	60

Table 4 Notes	
(a)	See **Section 3.3.3: Mulled Systems** for details
(b)	This does not apply to systems which are connected without the use of a mullion – for those systems, **Table 3: Forced Entry Testing of Individual Systems** applies

Table 5: Forced Entry Testing of Double Door Systems

Concentrated Assault Test	Specified Protection Level		
	5 Minutes	15 Minutes	60 Minutes
Required Testing Time (Minutes):			
Seam Between the Doors	5	15	60
Hinge of Active Door Leaf	5	15	60
Hinge of Inactive Door Leaf (a)	5	15	60
Additional Selected Locations (b)	5	15	60

Table 5 Notes	
(a)	This may be waived under **Section 3.1.1: Geometric Commonality and Physical Construction** at the discretion of DS/PSP/PSD representative
(b)	See **Section 3.3.2: Concentrated Assault Test**

Table 6: Minimum Ballistic Requirements [a]

Protection	Impact Locations		Ballistic Threat Caliber [1]	Fair Shots	Witness Panel
Rifle, Military (R)	Opaque Areas	Main Panel Body [2]	5.56 mm, M855 5.56 mm, M193 7.62 mm, M80	8 8 8	d d d
	Opaque Areas	Seams, Framing Connections, Etc. *	5.56 mm, M855 5.56 mm, M193 7.62 mm, M80	3 3 3	d d d
	Transparencies	Main Panel Body [2]	5.56 mm, M855 5.56 mm, M193 7.62 mm, M80	8 8 8	e e e
	Transparencies	Confined Area (i.e. corner) *	5.56 mm, M855 5.56 mm, M193 7.62 mm, M80	1 1 1	e e e
Rifle, Armor Piercing (AP)	Opaque Areas	Main Panel Body [3]	.30-06	4	d
	Opaque Areas	Seams, Framing Connections, Etc. *	.30-06	3	d
	Transparencies	Main Panel Body [3]	.30-06	4	e
	Transparencies	Confined Area (i.e. corner) *	.30-06	1	e
Shotgun (SH)	Unobstructed, Convoluted Passages (Deal Trays, Speak-Through Devices, Etc.) *		12 gauge 70mm #4 Buckshot	1	d

Witness Panel	Thickness		Hardness	Yield Strength	Description
	in.	mm	Brinell	psi	
d	0.020 in. (± 0.003 in.)	0.5 mm (± 0.076 mm)	120 Brinell	41,000 psi	High-Strength 2024T3 Aluminum (Al) Sheet
e	0.002 in. (± 0.0002 in.)	0.050 mm (± 0.005 mm)	20 Brinell	2,500 psi	1100 Aluminum (Al) Foil, Temper Annealed

Table 6 Notes	
a	Additional testing is at the Test Director's and DS/PSP/PSD Representative's option (**Section 3.2.2: Additional Firings**)
1	See **Table 1: Ballistic Resistance Test Ammunition**
2	See **Figure 3: 8-Shot Spacing**
3	See **Figure 4: 4-Shot Spacing**
*	Geometry will dictate the number of shots per approval from DS/PSP/PSD. Some examples of dissimilar areas are: Deal Trays, Doorknobs, Electrical Fittings, Hinges, HVAC Fittings, Irregular Angles, Locks, Louver Vanes, Mounting Gaps, Mullions, Oblique Angles (one shot for each identified oblique angle), and Speak-Through Devices. The preceding is not an exhaustive list.

Table 7: DOS Codes

Digit Position	Significance	Numbers
First X _ _ _	Composition	1 = Transparent 2 = Opaque
Second _ X _ _	Fire Resistance	1 = None 2 = 90 Minutes 3 = 180 Minutes 4 = 45 Minutes 5 = 20 Minutes 6 = 60 minutes
Third _ _ X _	FE Resistance (a)	1 = None 2 = 15 Minutes 3 = 60 Minutes 4 = 5 Minutes
Fourth _ _ _ X	Ballistic Resistance (b)	1 = None 3 = Rifle 4 = Armor Piercing
Example: A DOS code of “1123” would mean a transparent system, no fire rating, a 15-minute FE rating, and a Rifle BR rating.		

Table 7 Notes	
(a)	See **Section 1.3.1(a): Forced Entry Resistance Levels**
(b)	See **Section 1.3.2(a): Ballistic Resistance Levels**

Table 8 v3: Commercial Test Labs Approved by DOS to Conduct FE and BR Tests to SD-STD-01.01 (a)~~(b)~~(c)

Lab Name	Address	Contact Numbers	Contact Email
BRE Global	Bucknalls Lane, Garston Watford, Hertfordshire WD25 9XX	Tel: 44.0.1923.664000 Fax: 44.1923.664994	Richard Flint flintr@bre.co.uk
Element U.S. Space & Defense/EUSSD (formerly National Testing Service/NTS, Chesapeake Testing Services/CTS)	4603B Compass Point Road, Belcamp, MD 21017	Tel: 410-297-8154 Fax:410-297-8160	Matt Rixham matt.rixham@elementdefense.com
Intertek Laboratory (formerly Architectural Testing Inc./ATI)	130 Derry Court York, PA 17406	Tel: 717-764-7700 Fax: 717-764-4129	Travis Hoover Travis.Hoover@Intertek.com
Integrated Solutions for Systems, Inc./IS4S	2600 Partin Drive N Suite 220 Niceville, FL 32578	Tel: 850-428-3604	Brett Blazer brett.blazer@is4s.com
National Institute for Aviation Research/NIAR, Wichita State University	1845 Fairmount Street Wichita, KS 67260	Tel: 316-978-6427 Fax: 316-978-3175	James Dellinger jdellinger@niar.wichita.edu
Newtec Services	333 Hart Street P.O. Box 643 Edgefield, SC 29824	Tel: 803-480-0410	Keith Williams newtecsgi@gmail.com
Susquehanna Labs	3114 Scarboro Road MD, 21154	Tel: 443-347-6025 Mob: 443-347-5934 (Steve) Mob: 443-347-6027 (Mike)	Steve McDowell, Owner Steve@susqlabs.com Mike Hinder, Owner Mike@susqlabs.com
Oregon Ballistics Lab/OBL	2873 22nd Street SE Salem, OR 97302	Tel: 503-540-8114 Fax: 503-362-5597	Tom Ohnstad tohnstad@oregonbl.com Brandon Bertsch bbertsch@oregonbl.com

Table 8 v3 Notes	
(a)	This table is subject to change by DS/PSP/PSD
(b)	~~This version, **Table 8 v2** (dated October 24, 2024), supersedes **Table 8 v1** (dated June 16, 2020)~~
(c)	This version, Table 8 v3 (April 2026), supersedes Table 8 v2 (October 24, 2024)

TAB 3 PAGE 8 (v2)

Table 9: DOS Model Numbering Abbreviations for System Type (a)

Type	Description	Abbrev.
Booth	Freestanding	FB
	Regular	B
	DOS Opaque	BO
	DOS Transparent	BT
	Special	SP
Building	Modular Portable	BMP
Door	• Double	DD
	• Double Louvered	DDL
	• Double Louvered Opaque	DDLO
	• Double Opaque	DDO
	• Double Special	DDX
	• Double Transparent	DDT
	• Fire	DF
	• Louvered	DL
	• Opaque	DO
	• Special	DX
	• Transparent	DT
	• Vault	DV
	• Roller	DR
Gap	Regular	GP
Grill	Grate	GGL
	Regular	GL
	Special	GX
	Window	GW
Hatch	Escape	HE
Mullion	Regular	ML
Panel	Vision	PV
Partition	Combination	PK
	Concrete	PC
	Louvered	PL
	Masonry	PM
	Special	PX
	Steel	PS
	Transaction	PT
	Window	PW
	Vision	PW
	Opaque	PO
Screen	Mesh	SM
	Other	SO
Special Equipment	Enclosure	EE
	Gunport	SGP
	Shield	ES
	Other Purpose	SP
Tray	Deal Tray	TD
	Deal Drawer	TDD
Vault	Modular	MV
	Other	VO
Window	Skylight	SKY
	Speak-Through	WS
	Transaction	WT
	Vision	WV

Table 9 Notes	
(a)	Subject to change by DS/PSP/PSD

TAB 4: FORMS

Note that the following forms are to be filled out by the test facility:

- **Form 1: Test Facility Data**
- **Form 2: Pre-Test Checkoff List to Be Completed by Test Facility**
- **Form 3: Test Specimen Ballistic Data Records**
- **Form 5: Verification of Sample Form**
- **Form 6: Forced Entry Data Records**

Note that the following form is to be filled out by the manufacturer:

- **Form 4: FE/BR Certification Request Form**

Form 1: Test Facility Data

Test Number: ______________________________

Test Date: ______________________________

Test Time: ______________________________

Test Facility Name: ______________________________

Test Location (e.g. Building #): ____________________

Attack Area on Sample: ______________________________

Test Director Name: ______________________________

Test Team Personnel

	Team Member Name (First, M.I., Last)	Age on Day of Test (yrs.)	Weight (kg.)	Height (cm.)	Dominant Hand: R=Right, L=Left, R/L, or L/R	Team Member or Replacement?
1						
2						
3						
4						
5						
6						
7						

Microsoft Word version of this form available upon request from DS/PSD/RD. This data shall be recorded and included with the test report.

Form 2: Pre-Test Checkoff List to Be Completed by Test Facility

Sample Manufacturer: ______________________________

Manufacturer's Model Number: ______________________________

Describe Test Sample: ______________________________

Scheduled Test Date: ______________________________

Pre-Test Documentation Requirements

	Documentation	On File? (yes or no)	Date Received by Test Facility	Date Sent to DS/PSP/PSD
1	Installation Instructions per **Section 1.8.3: Installation Instructions** and **Section 2.2.1(a): Examination**			
2	Disclosure Documentation including Drawings & Specs. per **Section 1.8.2: Minimum Requirements for Engineering Drawings and Specifications** and **Section 2.2.1(a): Examination**			
3	OBO Specification Guarantee per **Section 1.8.4: OBO Specification Guarantee** and **Section 2.2.1(a): Examination**			
4	Signed Test-to-Failure Agreement per **Section 1.8.5: Test-to-Failure Agreement** and **Section 2.2.1(a): Examination**			
5	**Form 4: FE/BR Certification Request** Including: • Forced Entry Rating Sought per **Section 1.8.1: Request for Certification** and **Section 1.3.1(a): Forced Entry Resistance Levels** • Ballistic Resistance Rating Sought per **Section 1.8.1: Request for Certification** and **Section 1.3.2(a): Ballistic Resistance Levels**			
6	Formal Test Plan per **Section 2.2.1(b): Formal Test Plan**		N/A	
7	Written approval by DS/PSP/PSD of: • The Formal Test Plan • The Scheduled Test Date			N/A
8	**Form 5: Verification of Sample** Verification by test facility that the test sample was identical to the system shown in the manufacturer's submitted drawings per **Section 4.2.3(b): Verification of Identical System**		N/A	

The documents listed in this form have been reviewed and compared to the test sample and comply with the current version of SD-STD-01.01 with the following exceptions:

____

____

____

____

____

____

Microsoft Word version of **Form 2: Pre-Test Checkoff List to Be Completed by Test Facility** available upon request from DS/PSD/RD.

Form 3: Test Specimen Ballistic Data Records

Test Number: ______________________________

Test Date: ______________________________

Test Time: ______________________________

Ambient Temperature (°F): ____________________

Test Facility Name: __________________________

Test Location (e.g. Building #): _________________

Sample Manufacturer: ______________________________

Manufacturer's Model Number: ______________________

Test Specimen Identification Number: _______________________

Test Director Name: ______________________________

Test Director Signature:__________________________________**Date:** ____________

Observed By: ____

Ballistic Threat Ammunition

	Ammunition Description	Ammunition Identity	Bullet Weight (grain)	Minimum Velocity mps (fps)	Maximum Velocity mps (fps)	Ballistic Manufacturer & ID / Lot #	COTS or Hand Load?
1	7.62x51mm, NATO Ball, Steel Jacketed Lead Core, FMJ	M80	147 gr.	823.0 mps (2700 fps)	853.5 mps (2800 fps)		
2	5.56x45mm, M193, Copper Jacket Lead Core, FMJ	M193	55 gr.	955.5 mps (3135 fps)	986.0 mps (3235 fps)		
3	5.56x45mm, Copper Jacketed Steel and Lead Core, FMJ	M855	63 gr.	899.2 mps (2950 fps)	929.6 mps (3050 fps)		
4	.30-06 Caliber Rifle, Copper Jacketed Steel Core, APM2, (AP)	Original Ammunition Design	166 gr.	838.2 mps (2750 fps)	868.7 mps (2850 fps)		

mps = Meters Per Second, fps = Feet Per Second

Shots

	Shot Number	Threat	Velocity (fps)	Velocity (mps)	Bullet Spall?	Penetration NP = None PP = Partial CP = Complete	Unfair Hit?	Impact Location
1								
2								
3								
4								
5								

	Shot Number	Threat	Velocity (fps)	Velocity (mps)	Bullet Spall?	Penetration NP = None PP = Partial CP = Complete	Unfair Hit?	Impact Location
6								
7								
8								
9								
10								
11								
12								
13								
14								
15								
16								
17								
18								
19								
20								
21								
22								
23								
24								
25								
26								
27								
28								
29								
30								
31								
32								
33								
34								
35								
36								
37								
38								
39								

Remarks

Shot Number	Remarks

Microsoft Word version of **Form 3: Test Specimen Ballistic Data Records** available upon request from DS/PSD/RD.

Form 4: FE/BR Certification Request Form

Manufacturer's Request for Department of State

Forced Entry (FE) / Ballistic Resistance (BR) Certification

(insert picture or drawing)	(insert picture or drawing)
Previously Certified System(s) – If Applicable	**System Submitted for Certification**

Manufacturer Information	Manufacturer Name: Enter name Street Address: Enter street address City, State, Zip: Enter city, state, zip	Manufacturer POC Name: Enter name Email: Enter email address Tel.: Enter telephone #

Product Information		Product type (general): Select general type Product type (specific): Select specific type	Specs. As Tested Size (inches): Enter size Weight (lbs.): Enter weight Material: Enter material Mounting: Enter mounting Mulled System (Y/N): Select Y/N Stacked System (Y/N): Select Y/N
Manufacturer's Internal Model Number(s): • ____________________	Optional Notes (i.e. custom options or exceptions): • ____________________		

Certification Request	Certification Mode Requested (Testing or T/R): Select mode
FE Rating Sought (0, 5, 15, or 60): Select FE BR Rating Sought (None, Rifle, or AP): Select BR Fire Rating Sought (0, 20, 45, 60, 90, or 180): Select fire Mulling Sought with (if Applicable, list system(s)): • ____________________	New DOS Model Number(s) Requested: • ____________________ Reason for Request for Certification: • ____________________

Table of Model Numbers

Description of System	Previous Manufacturer's Internal Model Number (if applicable)	New Manufacturer's Internal Model Number	Previous DOS Certification Letter Date (if applicable)	Previous DOS Model Number (if applicable)	New DOS Model Number Requested	New DOS Code

Documents

Select "Yes" if Submitted and Provide Date:

Letter of Intent requesting certification, and providing FE/BR testing level spec(s): Select Y/N MM/##/YYYY

Disclosure Documentation (engineering drawings): Select Y/N MM/##/YYYY

Installation Instructions: Select Y/N MM/##/YYYY

OBO Spec. Guarantee: Select Y/N MM/##/YYYY

Test-to-Failure Agreement: Select Y/N or N/A MM/##/YYYY

Report(s) for FE/BR, Cycle, and Fire Testing (UL):

1.Report #: Enter report number

a.Test Date: MM/##/YYYY

b.Test Purpose: Enter purpose

c. Report Date: MM/##/YYYY

d.Test Facility: Enter facility

2.Report #: Enter report number

a. Test Date: MM/##/YYYY

b.Test Purpose: Enter purpose

c. Report Date: MM/##/YYYY

d.Test Facility: Enter facility

...

(insert picture or drawing)	(insert picture or drawing)
Additional Image #1: Caption	**Additional Image #2: Caption**

Form 5: Verification of Sample Form

Test Number: ______________________________

Scheduled Test Date: ________________________

Sample Manufacturer: _____________________________

Manufacturer's Model Number: ______________________

Describe Test Sample: _____________________________

I,________________________, of________________________certify that the test sample submitted for testing by________________________is identical to the disclosure documentation, including drawings and specifications, submitted by________________________in accordance with ***Section 1.8.2: Minimum Requirements for Engineering Drawings*** *and* ***Specifications and Section 2.2.1(a): Examination****. I,________________________, of________________________, further certify that the test sample submitted for Forced Entry / Ballistic Resistance (FE/BR) testing was installed according to the installation instructions supplied by________________________in accordance with* ***Section 1.8.3: Installation Instructions*** *and* ***Section 2.2.1(a): Examination****.*

Test Facility Name: __________________________

Test Director Name: ______________________________

Test Director Signature:__________________________________**Date:** ____________

This form shall be included with the test report.

Form 6: Forced Entry Data Records

Test Number: ______________________________

Test Date: ______________________________

Test Time: ______________________________

Ambient Temperature (°F): ____________________

Test Facility Name: __________________________

Test Location (e.g. Building #): _________________

Attack Area on Sample: ______________________

Sample Manufacturer: ______________________________

Manufacturer's Model Number: _______________________

Test Specimen Identification Number: _______________________

Test Director Name: ______________________________

Test Director Signature:__________________________________**Date:** ____________

Observed By: ____

____

Attack

Time Interval	Written Narrative of Attack Progress
5 Minutes	
10 Minutes	
15 Minutes	
30 Minutes	
45 Minutes	
60 Minutes	
Time at Failure	
Provide narrative commentary at intervals above, as well as when there are significant changes in the test samples. Record the time(s) at which there are significant changes in the test sample. For 60-minute testing, take attack and protected side photos at each rest period.	

Microsoft Word version of **Form 6: Forced Entry Data Records** available upon request from DS/PSD/RD.